

Banana River Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

Contents

Funding & Acknowledgements	2
Threshold Filtering	2
Value Qualifiers	3
Water Column	5
Seasonal Kendall-Tau Analysis	5
Water Quality - Discrete	5
Chlorophyll a, Corrected for Pheophytin - Discrete	6
Chlorophyll a, Uncorrected for Pheophytin - Discrete	7
Colored Dissolved Organic Matter - Discrete	9
Dissolved Oxygen - Discrete	11
Dissolved Oxygen Saturation - Discrete	14
pH - Discrete	15
Salinity - Discrete	18
Total Nitrogen - Discrete	20
Total Phosphorus - Discrete	22
Turbidity - Discrete	24
Water Temperature - Discrete	26
Water Quality - Continuous	30
pH - Continuous	32
Dissolved Oxygen Saturation - Continuous	33
Dissolved Oxygen - Continuous	34
Turbidity - Continuous	35
Water Temperature - Continuous	36
Salinity - Continuous	37
Specific Conductivity - Continuous	38
Submerged Aquatic Vegetation	39
Parameters	39
Species	39
Notes	39
SAV Water Column Analysis	44
Species list	46
References	47

Funding & Acknowledgements

The data used in this analysis is from the Export Standardized Tables in the SEACAR Data Discovery Interface (DDI). Documents and information available through the SEACAR DDI are owned by the data provider(s) and users are expected to provide appropriate credit following accepted citation formats. Users are encouraged to access data to maximize utilization of gained knowledge, reducing redundant research and facilitating partnerships and scientific innovation.

With respect to documents and information available from SEACAR DDI, neither the State of Florida nor the Florida Department of Environmental Protection makes any warranty, expressed or implied, including the warranties of merchantability and fitness for a particular purpose arising out of the use or inability to use the data, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of any information, apparatus, product, or process disclosed, or represents that its use would not infringe privately owned rights.

This report was funded in part, through a grant agreement from the Florida Department of Environmental Protection, Florida Coastal Management Program, by a grant provided by the Office for Coastal Management under the Coastal Zone Management Act of 1972, as amended, National Oceanic and Atmospheric Administration. The views, statements, findings, conclusions and recommendations expressed herein are those of the author(s) and do not necessarily reflect the views of the State of Florida, NOAA or any of their sub agencies.

Published: 2026-06-10



Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m ⁻¹	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

<i>SEACAR QAQC Description</i>	<i>Include</i>	<i>SEACAR QAQCFlagCode</i>
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>MDL</i>	<i>Description</i>
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>Description</i>
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Corrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

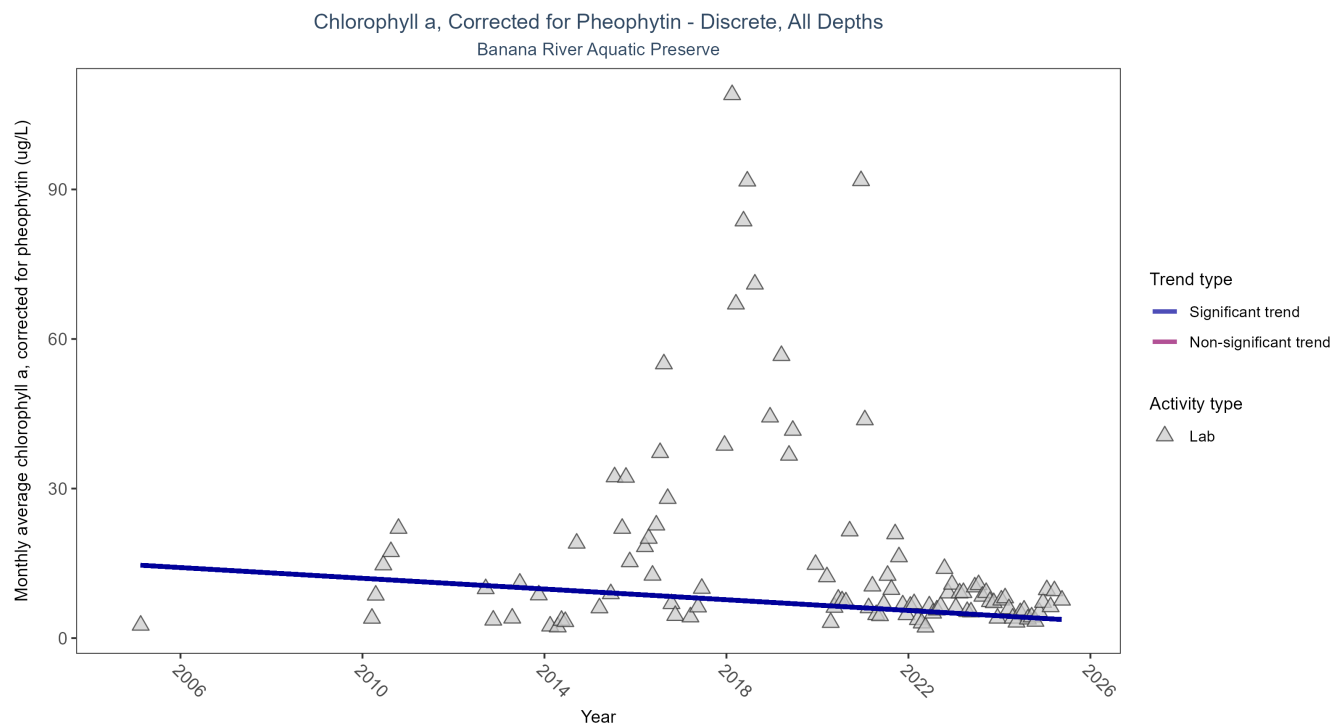


Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	555	16	2005 - 2025	6.4614	-0.2369	14.6885	-0.5379	0.0046

Monthly average chlorophyll a, corrected for pheophytin, decreased by 0.54 $\mu\text{g/L}$ per year, indicating an increase in water clarity.

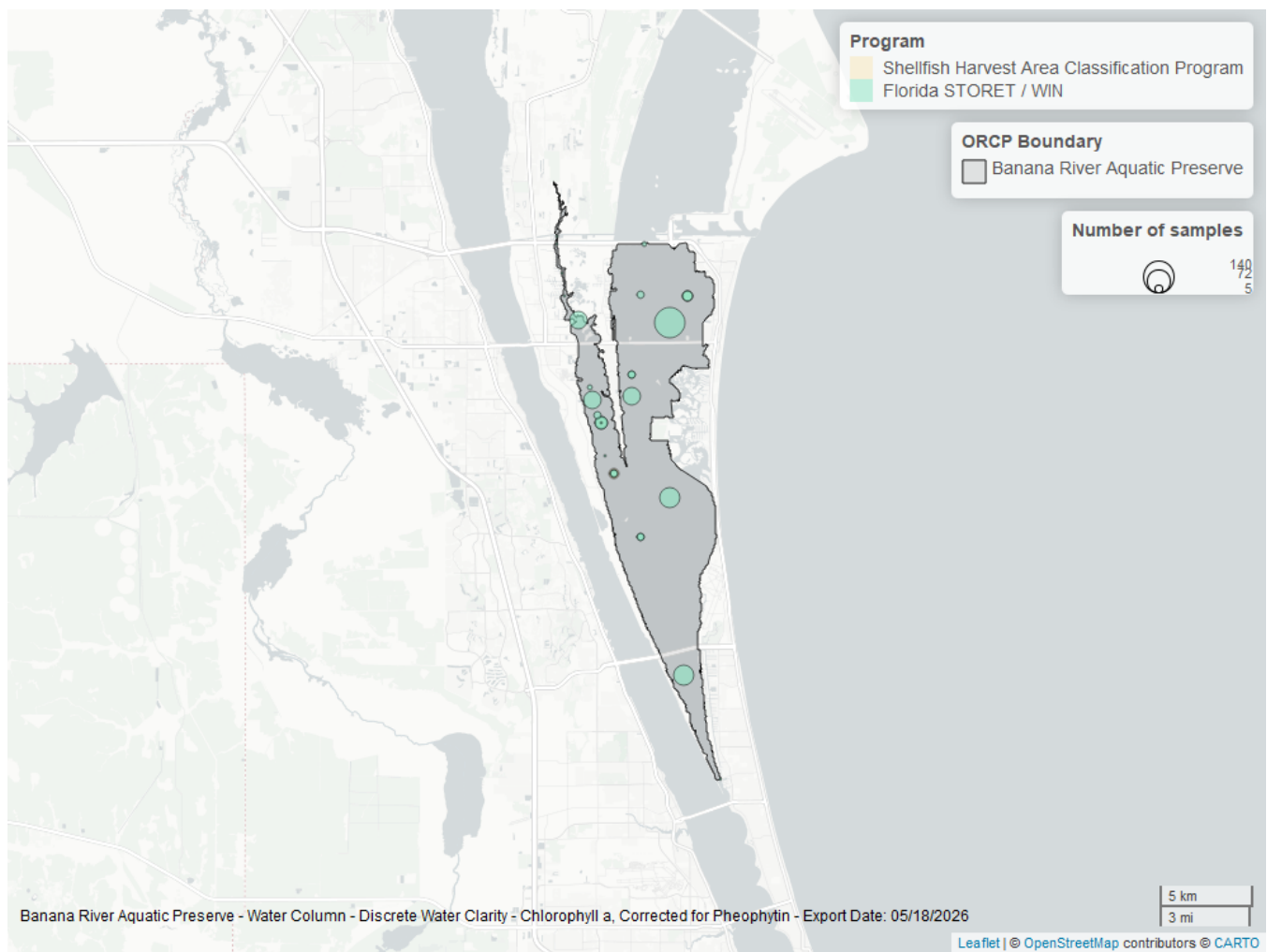


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5002	534	2005	2025
540	42	2016	2020

Program names:

5002 - Florida STORET / WIN¹

540 - Shellfish Harvest Area Classification Program²

Chlorophyll a, Uncorrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

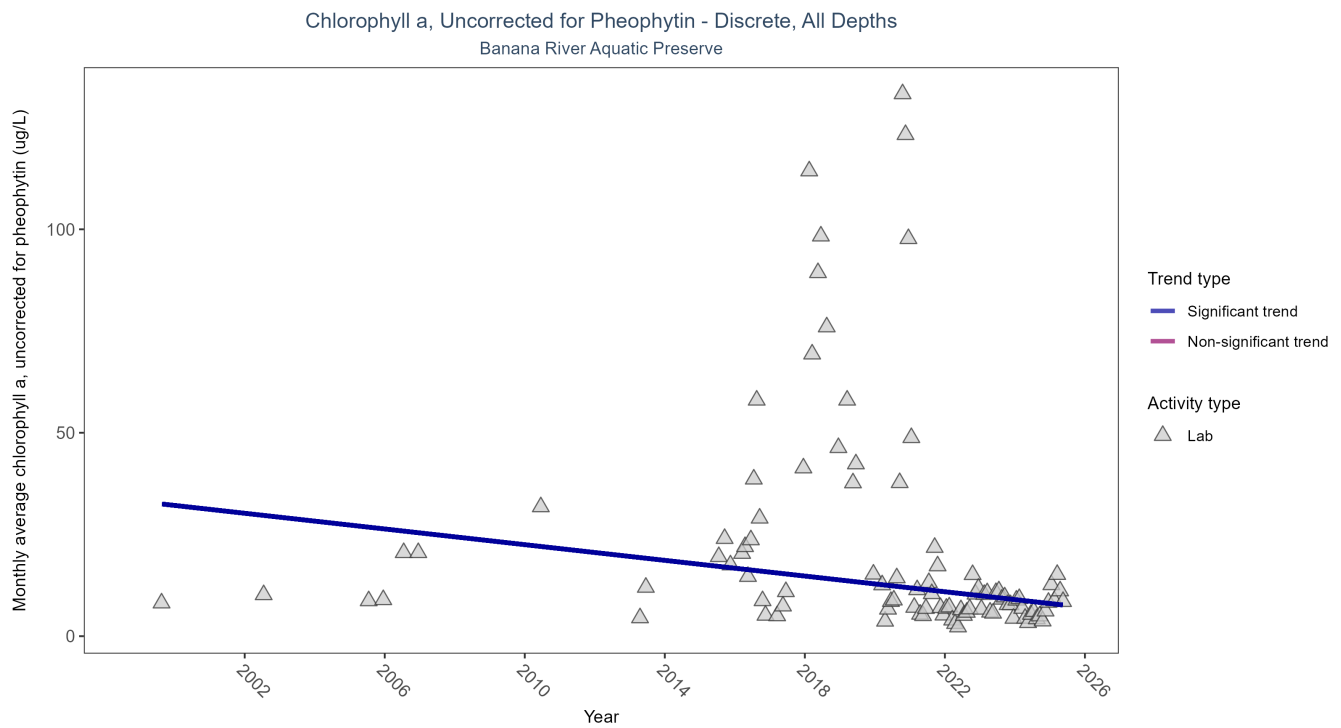


Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	519	17	1999 - 2025	7.6008	-0.3273	33.101	-0.9643	0

Monthly average chlorophyll a, uncorrected for pheophytin, decreased by 0.96 µg/L per year, indicating an increase in water clarity.

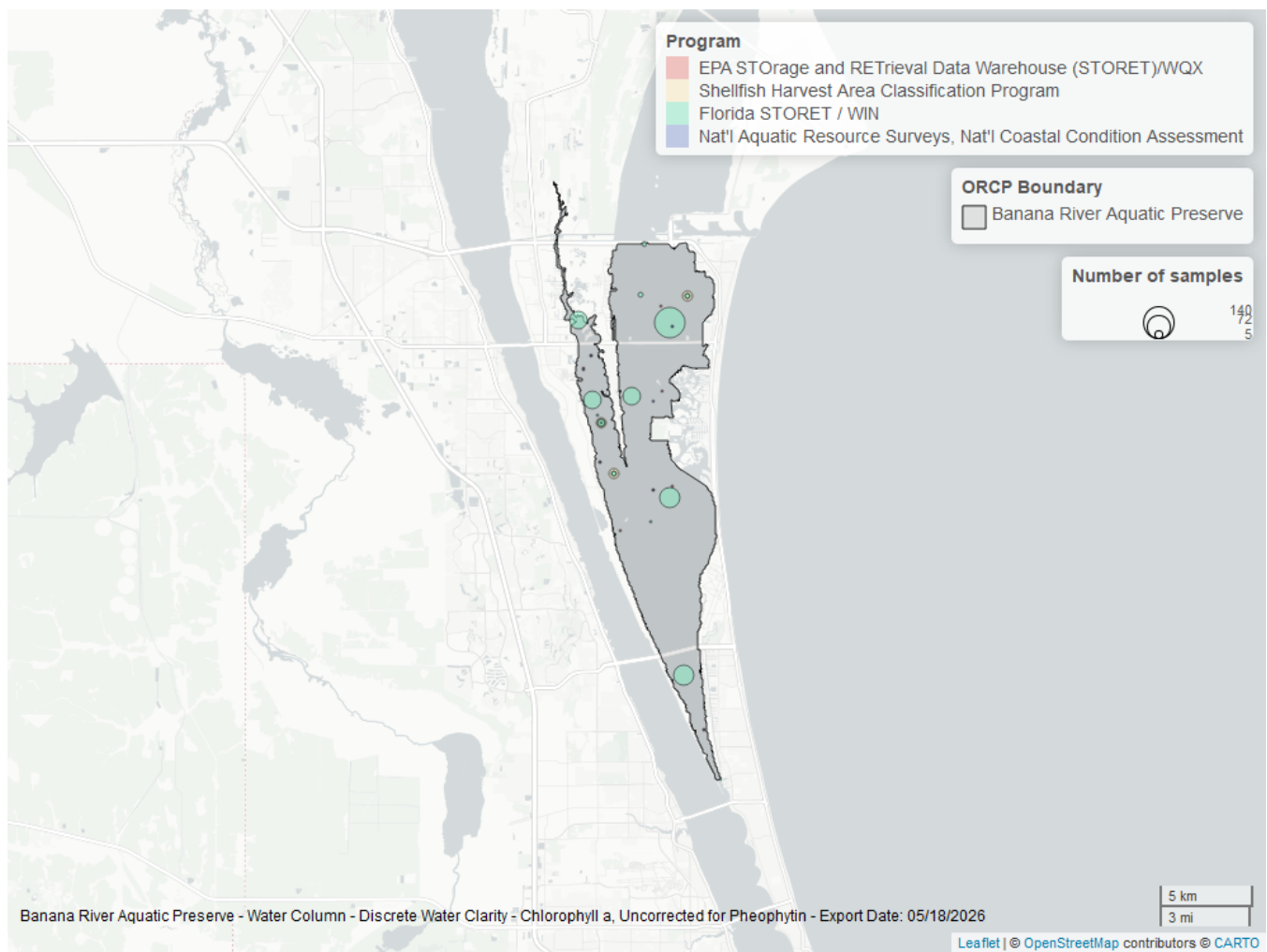


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
5002	458	1999	2025
540	42	2016	2020
103	13	2002	2015
118	8	2005	2010

Program names:

5002 - Florida STORET / WIN¹

540 - Shellfish Harvest Area Classification Program²

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX³

118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁴

Colored Dissolved Organic Matter - Discrete

Seasonal Kendall-Tau Trend Analysis

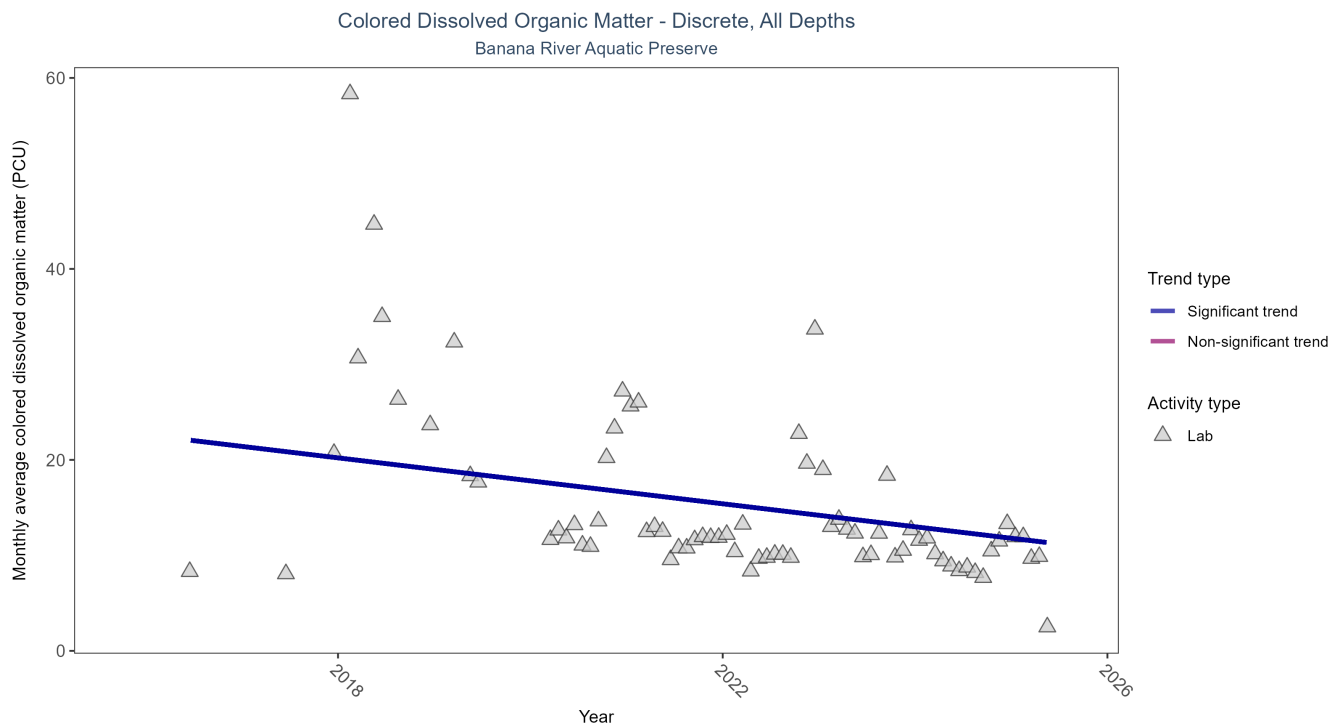


Figure 5: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Colored Dissolved Organic Matter

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	456	10	2016 - 2025	11.33	-0.4496	22.6161	-1.2025	0

Monthly average colored dissolved organic matter decreased by 1.2 PCU per year, indicating an increase in water clarity.

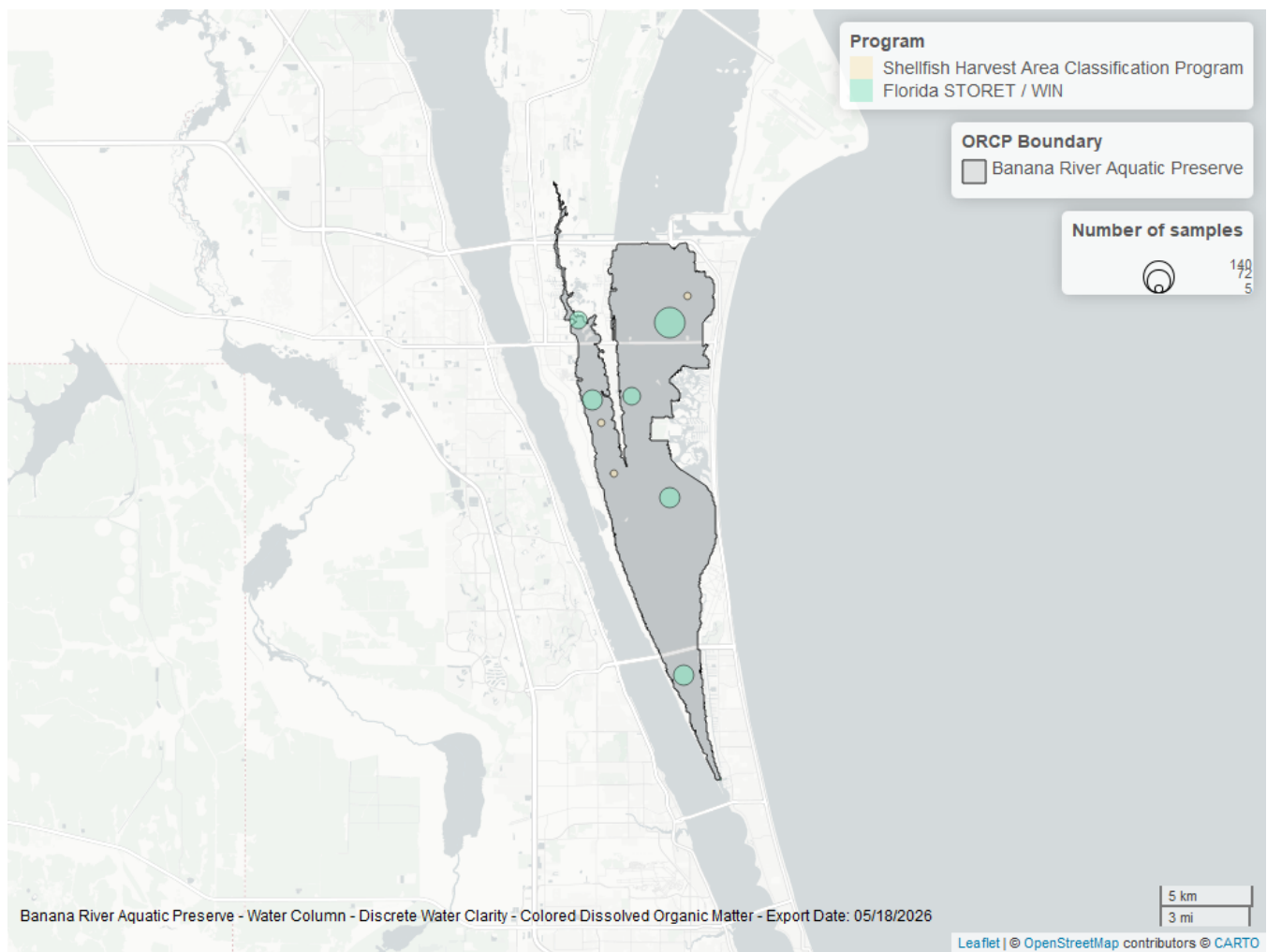


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Colored Dissolved Organic Matter

ProgramID	N_Data	YearMin	YearMax
5002	423	2020	2025
540	36	2016	2019

Program names:

5002 - Florida STORET / WIN¹

540 - Shellfish Harvest Area Classification Program²

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

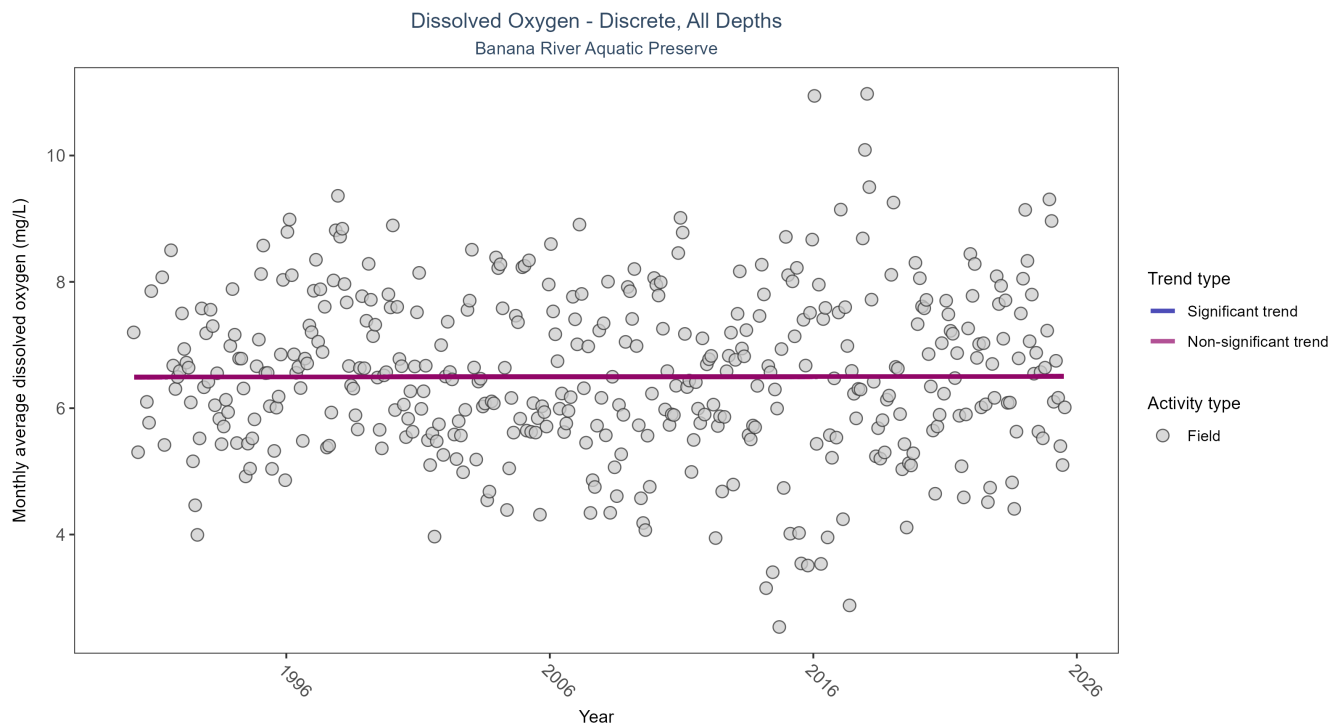


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	30082	36	1990 - 2025	6.5	0.0049	6.4932	0.0003	0.9203

Dissolved oxygen showed no detectable trend between 1990 and 2025.

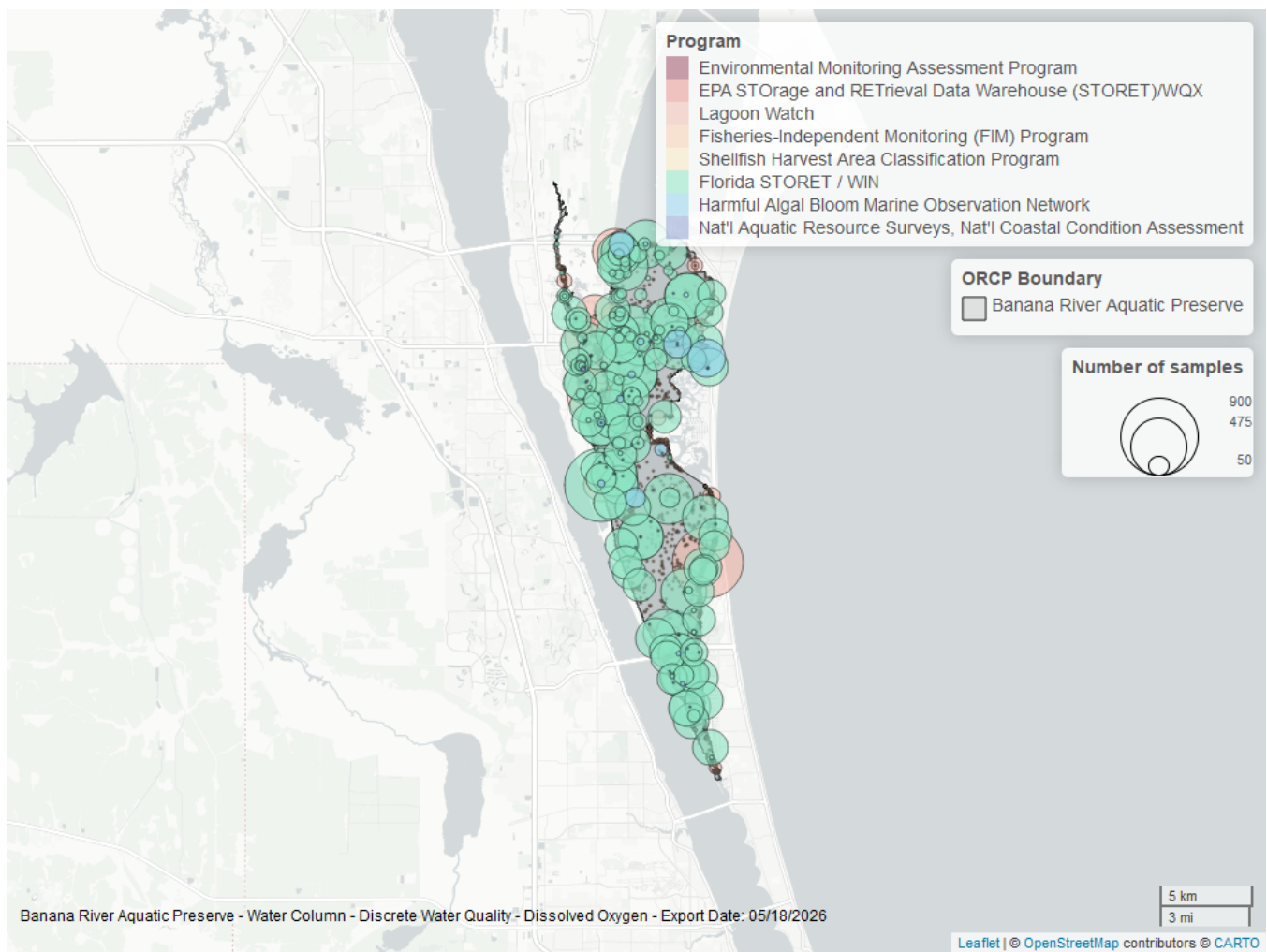


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
5002	20398	1991	2025
69	4530	1990	2024
3001	4249	1991	2023
95	658	2006	2018
3013	347	2003	2025
540	42	2016	2020
118	17	2005	2020
115	8	1995	1995
103	5	2015	2015

Program names:

5002 - Florida STORET / WIN¹

69 - Fisheries-Independent Monitoring (FIM) Program⁵

3001 - Lagoon Watch⁶

95 - Harmful Algal Bloom Marine Observation Network⁷

3013 - Indian River Lagoon fixed seagrass transects (SJRWMD)⁸
 540 - Shellfish Harvest Area Classification Program²
 118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁴
 115 - Environmental Monitoring Assessment Program⁹
 103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX³

Dissolved Oxygen Saturation - Discrete

Seasonal Kendall-Tau Trend Analysis

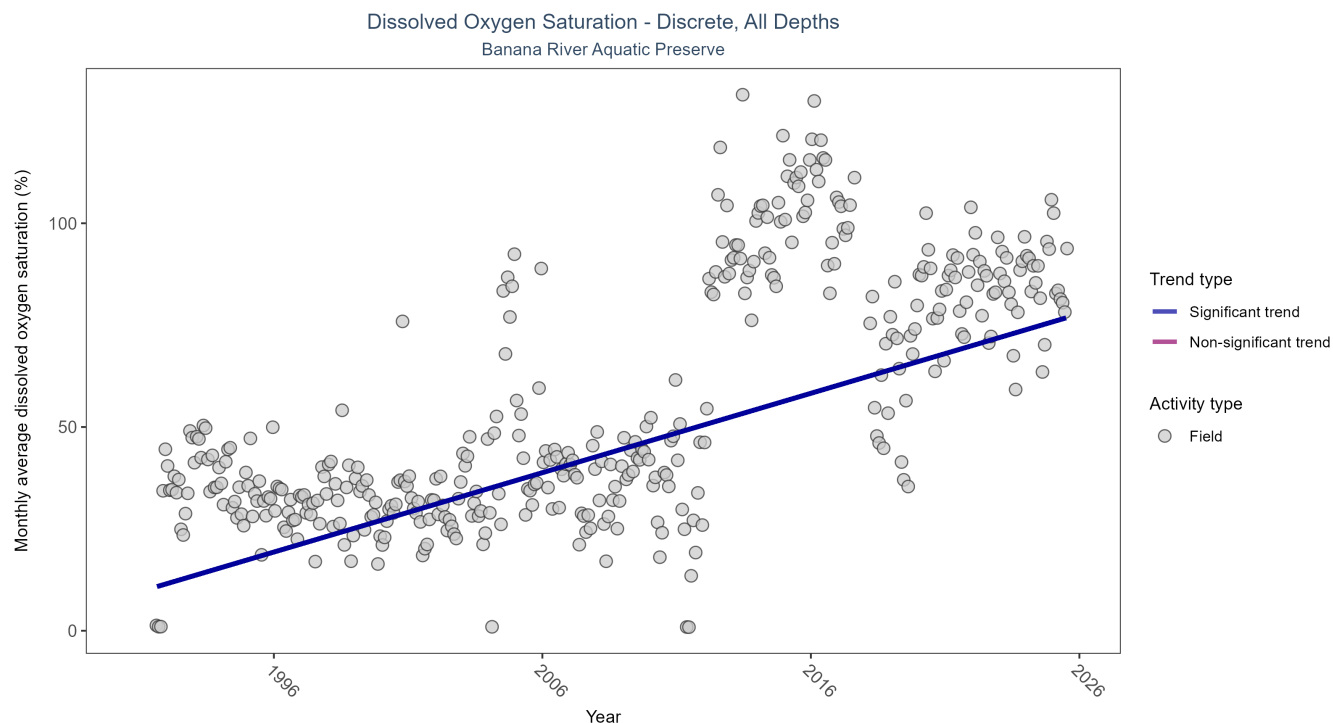


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	7670	35	1991 - 2025	60	0.4606	9.5731	1.948	0

Monthly average dissolved oxygen saturation increased by 1.95% per year.

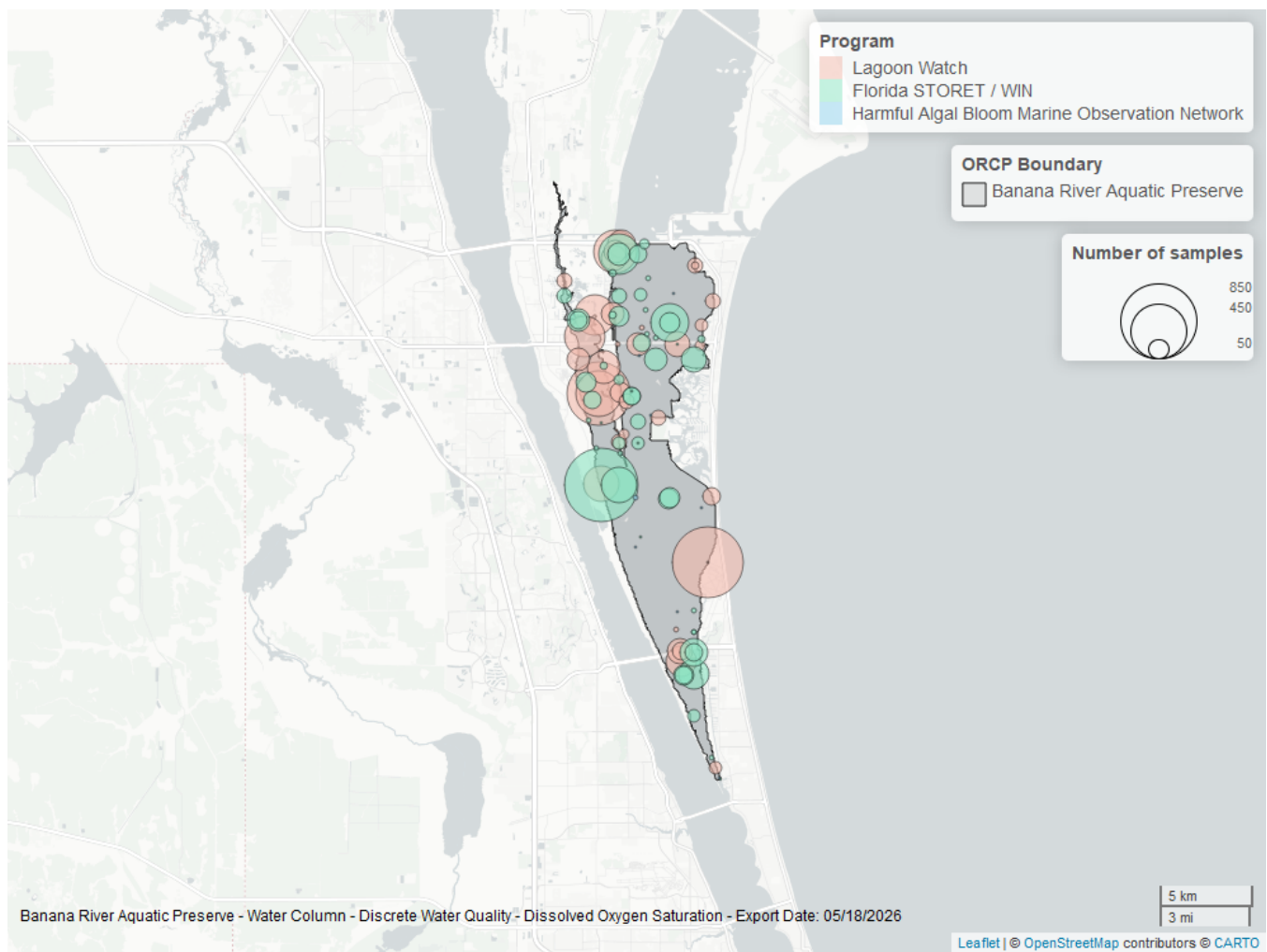


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Dissolved Oxygen Saturation

ProgramID	N_Data	YearMin	YearMax
3001	4238	1991	2024
5002	3262	1991	2025
3013	181	2012	2025
95	11	2014	2018

Program names:

3001 - Lagoon Watch⁶

5002 - Florida STORET / WIN¹

3013 - Indian River Lagoon fixed seagrass transects (SJRWMD)⁸

95 - Harmful Algal Bloom Marine Observation Network⁷

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

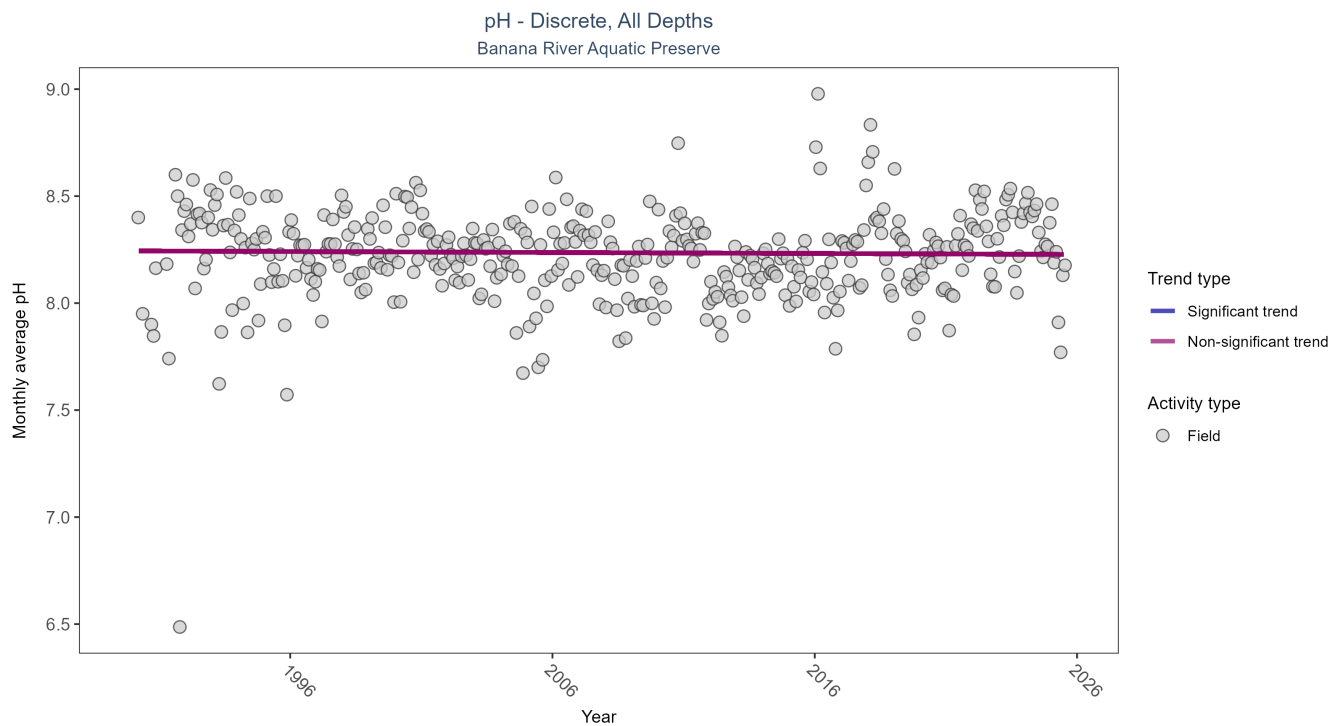


Figure 11: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	23027	36	1990 - 2025	8.2	-0.0189	8.244	-0.0005	0.6285

pH showed no detectable trend between 1990 and 2025.

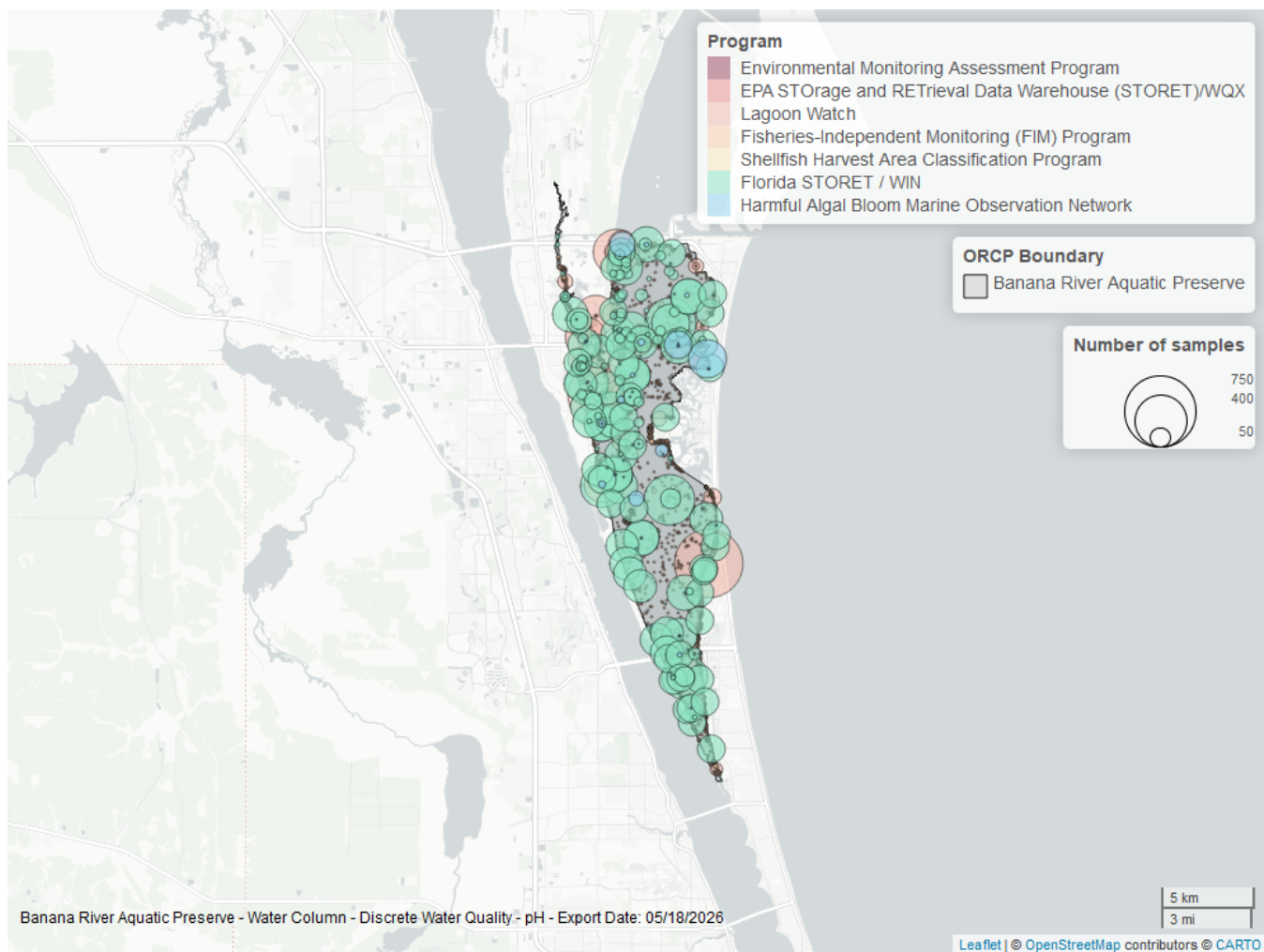


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
5002	13890	1996	2025
69	4493	1990	2024
3001	4169	1991	2024
95	615	2006	2018
3013	346	2003	2025
540	42	2016	2020
115	7	1995	1995
103	2	2015	2015

Program names:

5002 - Florida STORET / WIN¹

69 - Fisheries-Independent Monitoring (FIM) Program⁵

3001 - Lagoon Watch⁶

95 - Harmful Algal Bloom Marine Observation Network⁷

3013 - Indian River Lagoon fixed seagrass transects (SJRWMD)⁸

540 - Shellfish Harvest Area Classification Program²
 115 - Environmental Monitoring Assessment Program⁹
 103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX³

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

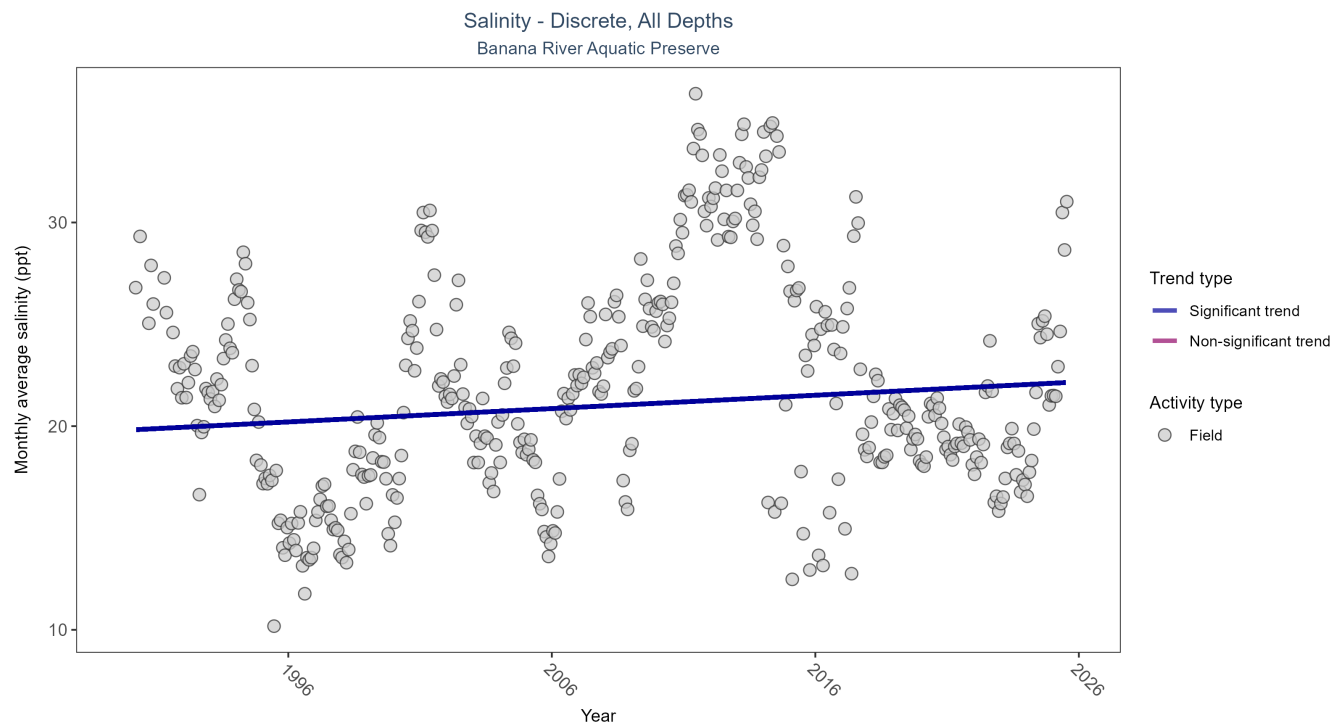


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Increasing trend	31765	36	1990 - 2025	19.8	0.0868	19.8134	0.0655	0.0123

Monthly average salinity increased by 0.07 ppt per year.

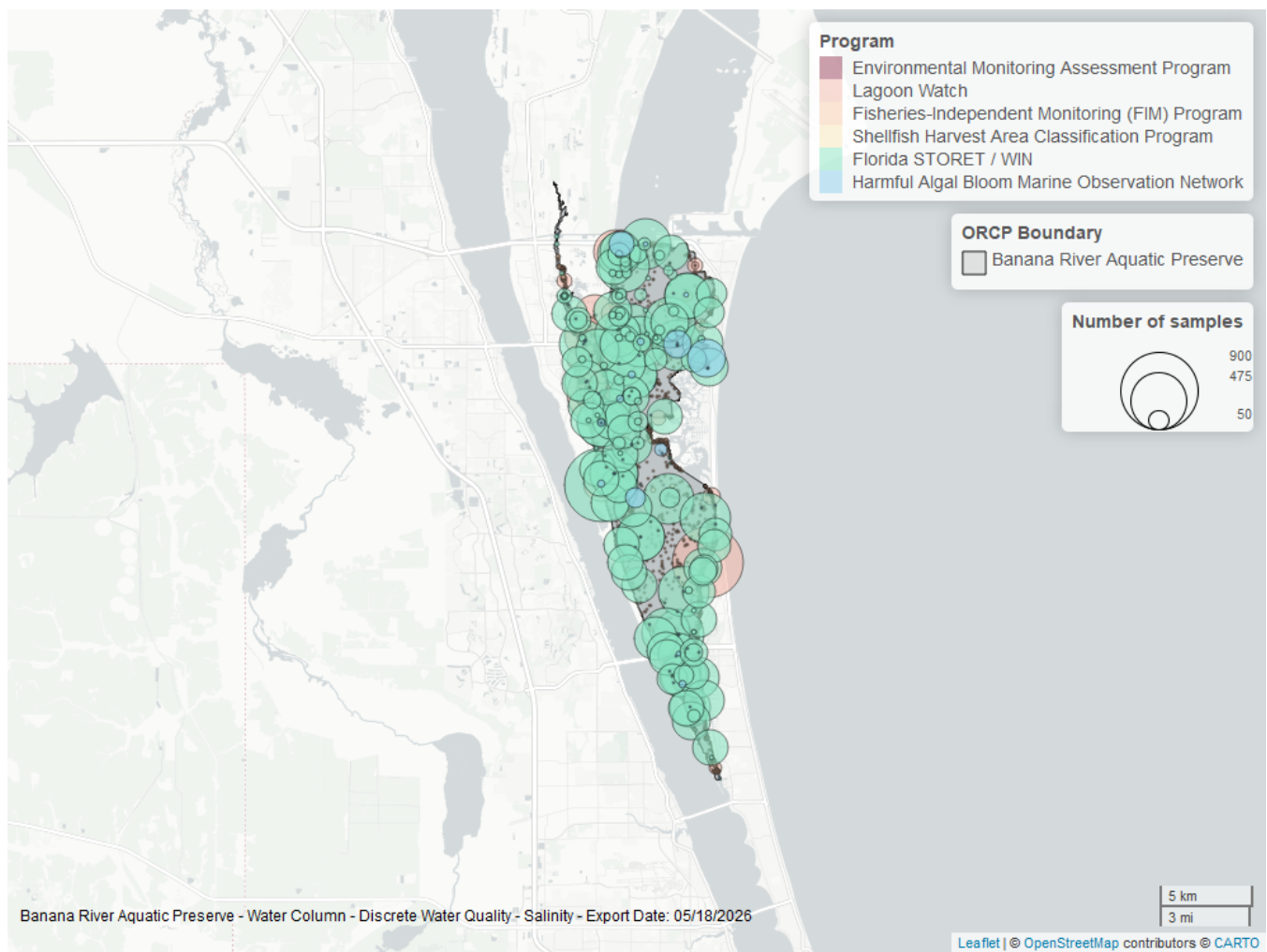


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
5002	21860	1991	2025
69	4536	1990	2024
3001	4317	1991	2024
95	670	2006	2018
3013	348	2003	2025
540	42	2016	2020
115	6	1995	1995

Program names:

5002 - Florida STORET / WIN¹

69 - Fisheries-Independent Monitoring (FIM) Program⁵

3001 - Lagoon Watch⁶

95 - Harmful Algal Bloom Marine Observation Network⁷

3013 - Indian River Lagoon fixed seagrass transects (SJRWMD)⁸

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

- 1) $TN = TKN + NO3O2$;
- 2) $TN = TKN + NO3 + NO2$;
- 3) $TN = ORGN + NH4 + NO3O2$;
- 4) $TN = ORGN + NH4 + NO2 + NO3$;
- 5) $TN = TKN + NO3$;
- 6) $TN = ORGN + NH4 + NO3$;

Additional Information:

- Rules for use of sample fraction:
 - Florida Department of Environmental Protection (FDEP) report that if both “Total” and “Dissolved” components are reported, only “Total” is used. If the total is not reported, then the dissolved components are used as a best available replacement.
 - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
 - ParameterName = “Total Nitrogen”
 - SEACAR_QAQCFlagCode = “1Q”
 - SEACAR_QAQC_Description = “SEACAR Calculated”

Seasonal Kendall-Tau Trend Analysis

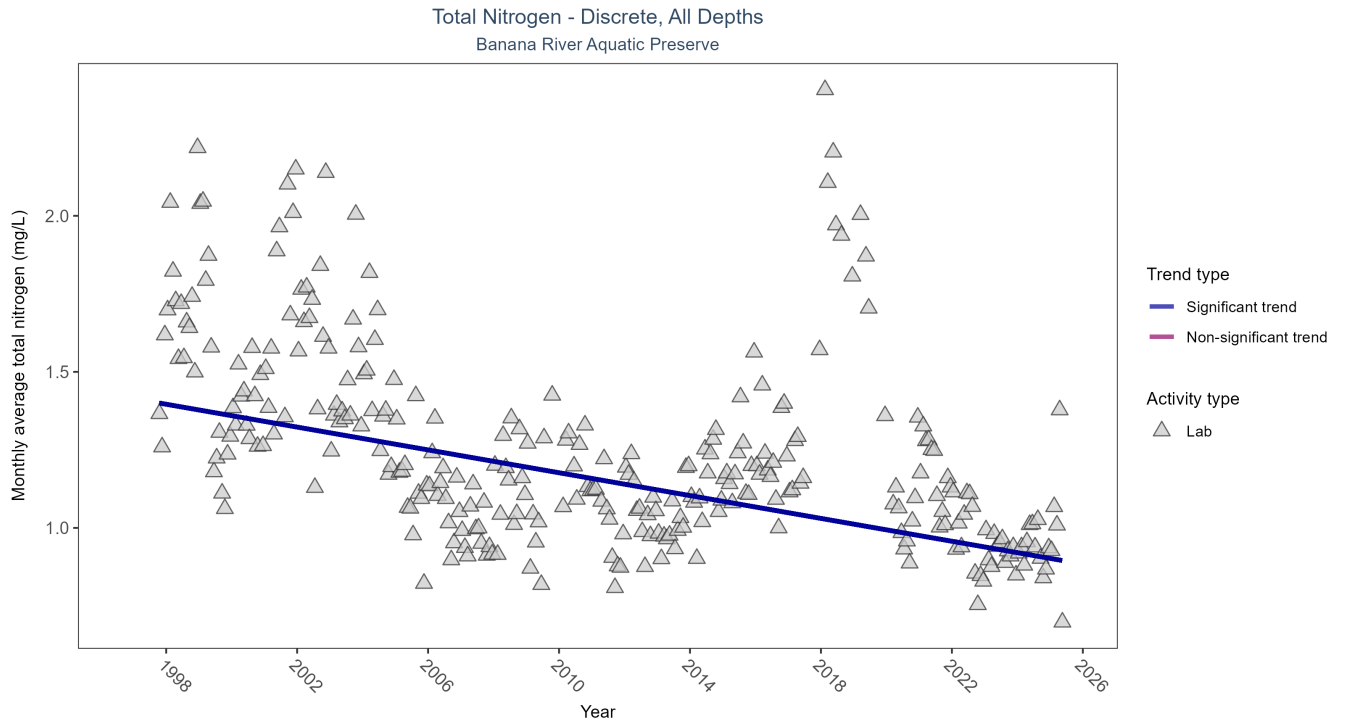


Figure 15: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	2336	29	1997 - 2025	1.2065	-0.4147	1.4145	-0.0183	0

Monthly average total nitrogen decreased by 0.02 mg/L per year.

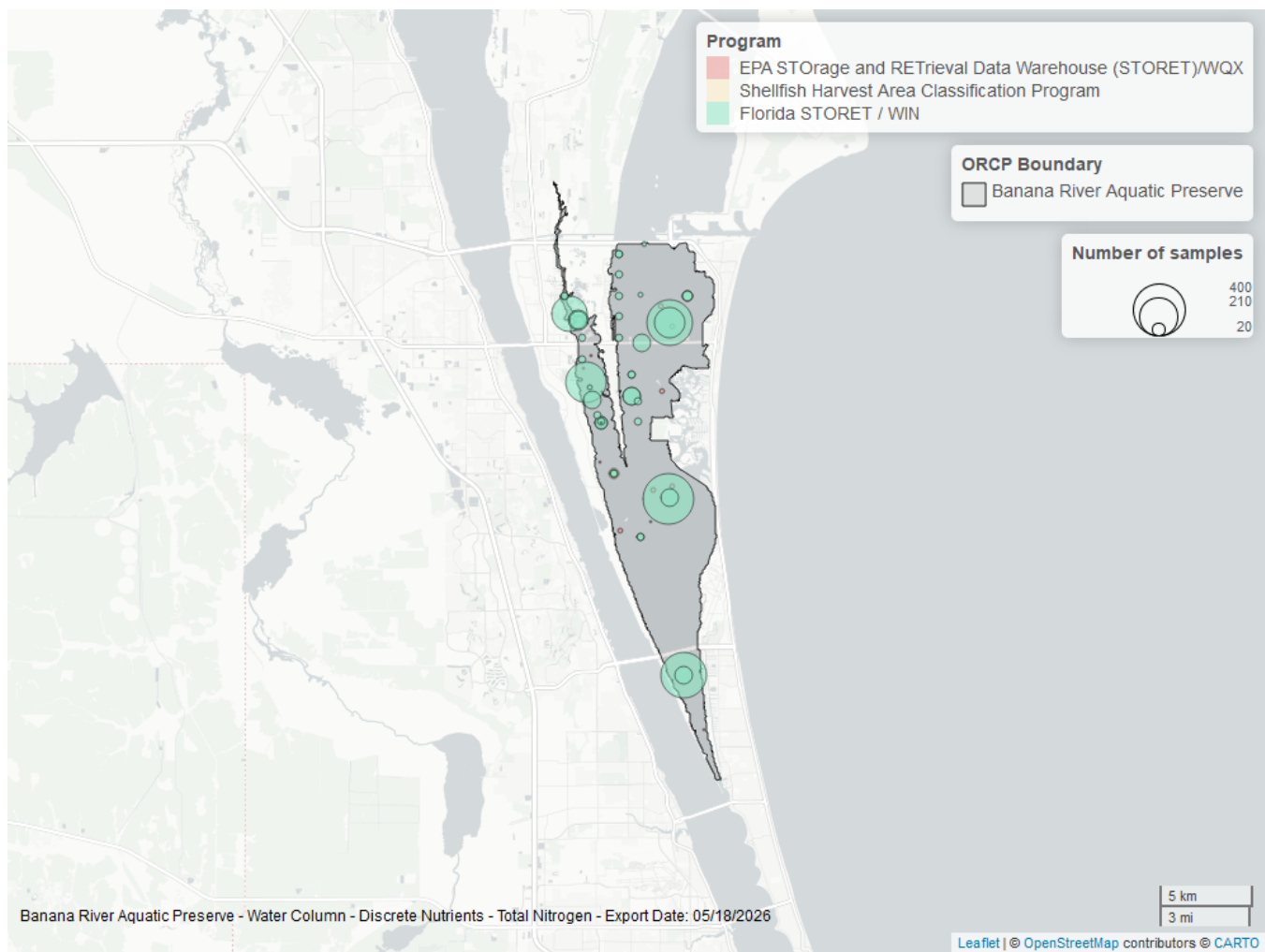


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
5002	2265	1997	2025
540	42	2016	2020
103	29	2002	2015

Program names:

5002 - Florida STORET / WIN¹

540 - Shellfish Harvest Area Classification Program²

103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX³

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

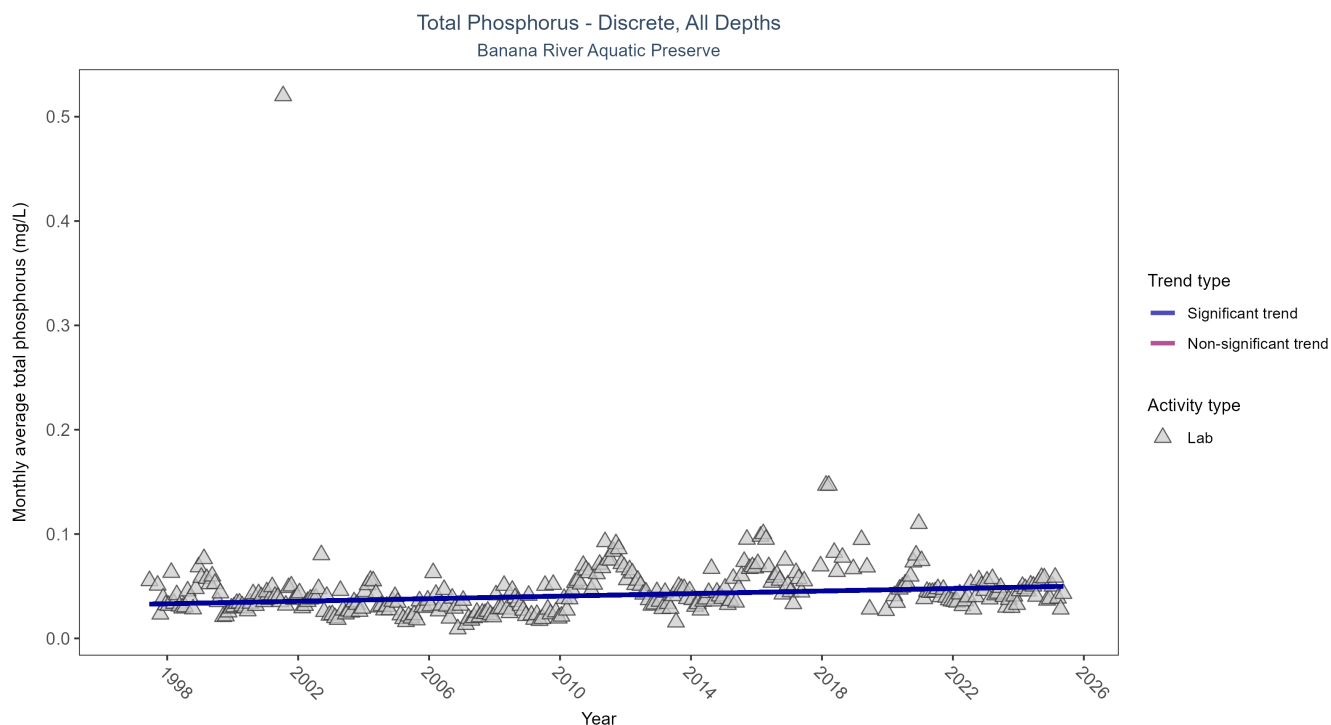


Figure 17: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	4827	29	1997 - 2025	0.0356	0.2241	0.0326	0.0006	0

Monthly average total phosphorus increased by less than 0.01 mg/L per year.

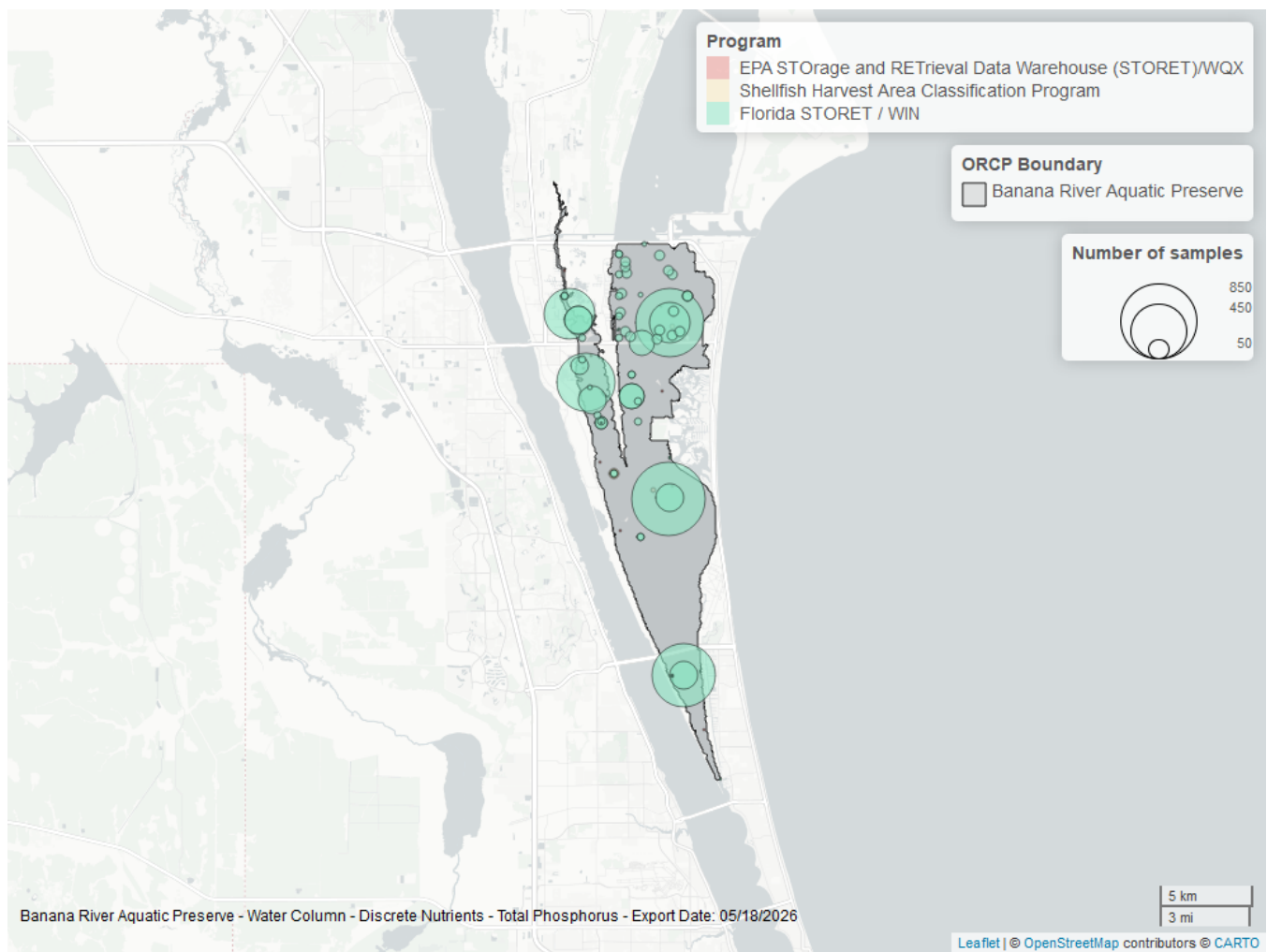


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
5002	4794	1997	2025
540	42	2016	2020
103	25	2002	2015

Program names:

5002 - Florida STORET / WIN¹

540 - Shellfish Harvest Area Classification Program²

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX³

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

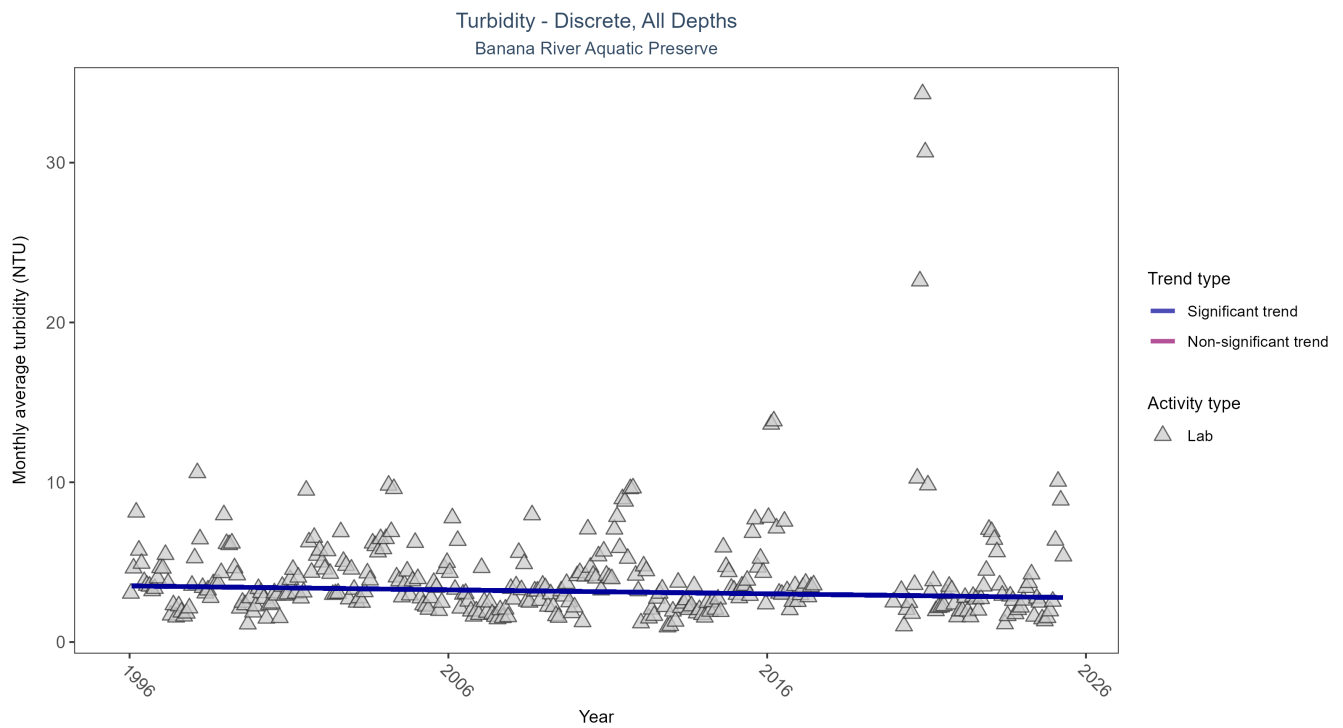


Figure 19: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	13558	29	1996 - 2025	3.1382	-0.101	3.5139	-0.0248	0.011

Monthly average turbidity decreased by 0.02 NTU per year, indicating an increase in water clarity.

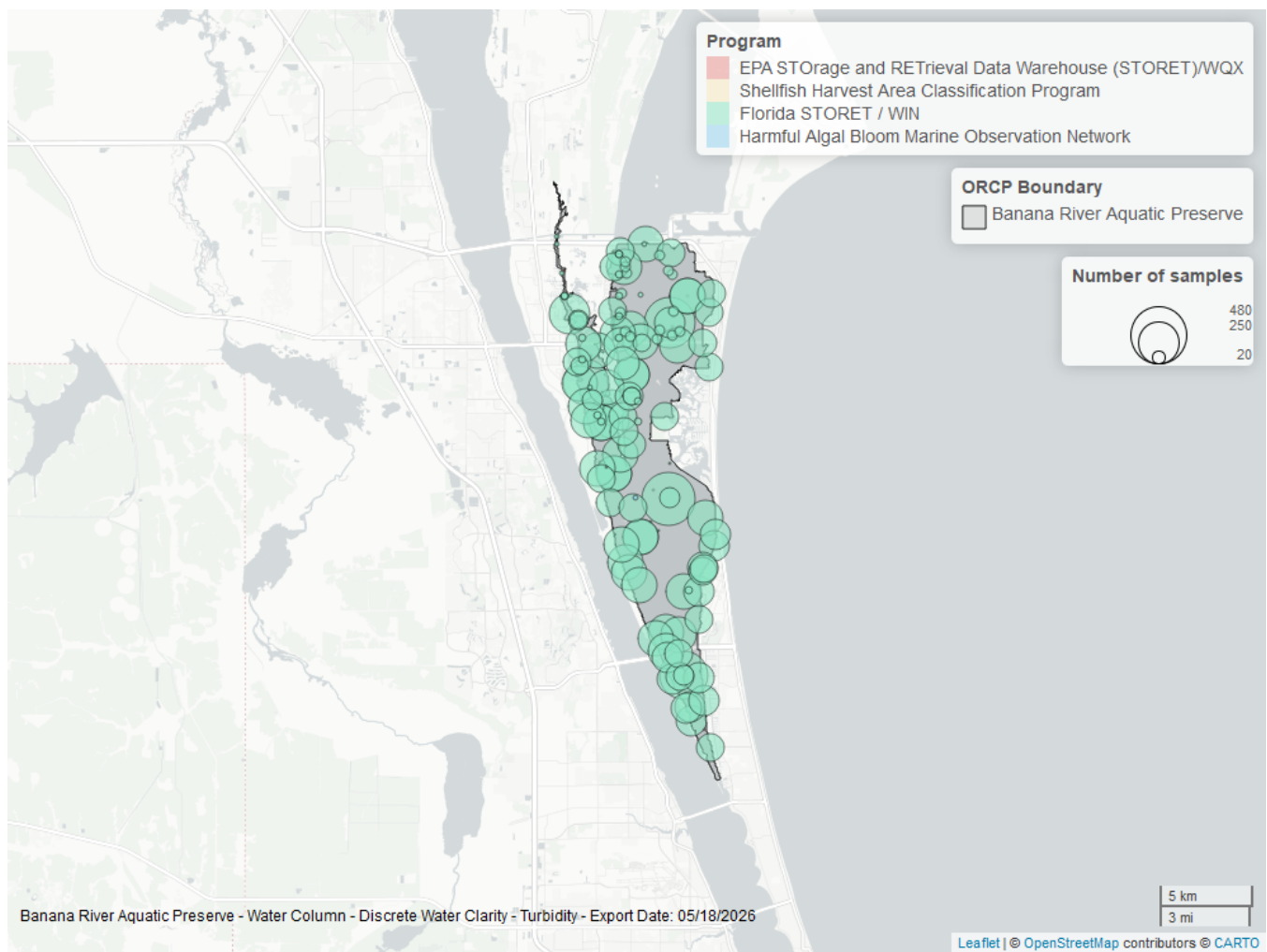


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5002	13787	1996	2025
3013	271	2004	2019
103	8	2005	2006
540	6	2019	2020
95	4	2009	2010

Program names:

5002 - Florida STORET / WIN¹

3013 - Indian River Lagoon fixed seagrass transects (SJRWMD)⁸

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX³

540 - Shellfish Harvest Area Classification Program²

95 - Harmful Algal Bloom Marine Observation Network⁷

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

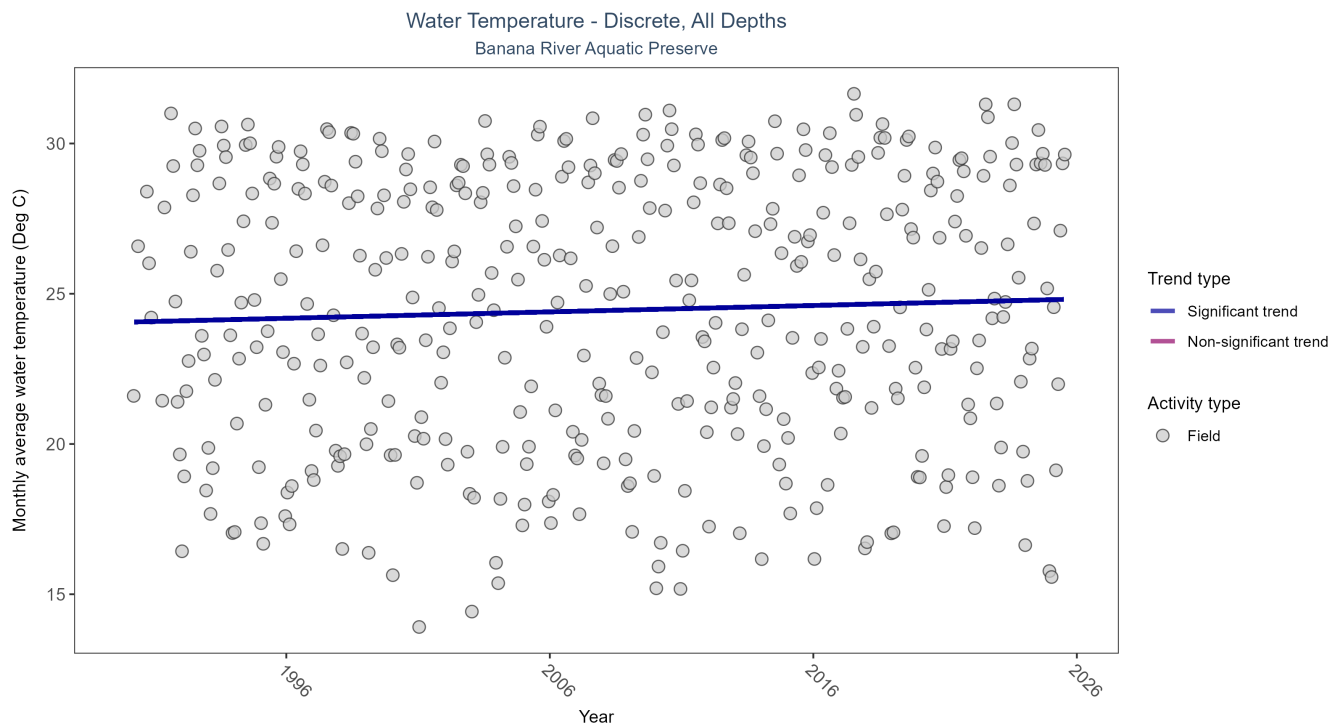


Figure 21: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 26: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	31615	36	1990 - 2025	25.5	0.1208	24.0557	0.0213	0.0004

Monthly average water temperature increased by 0.02°C per year.

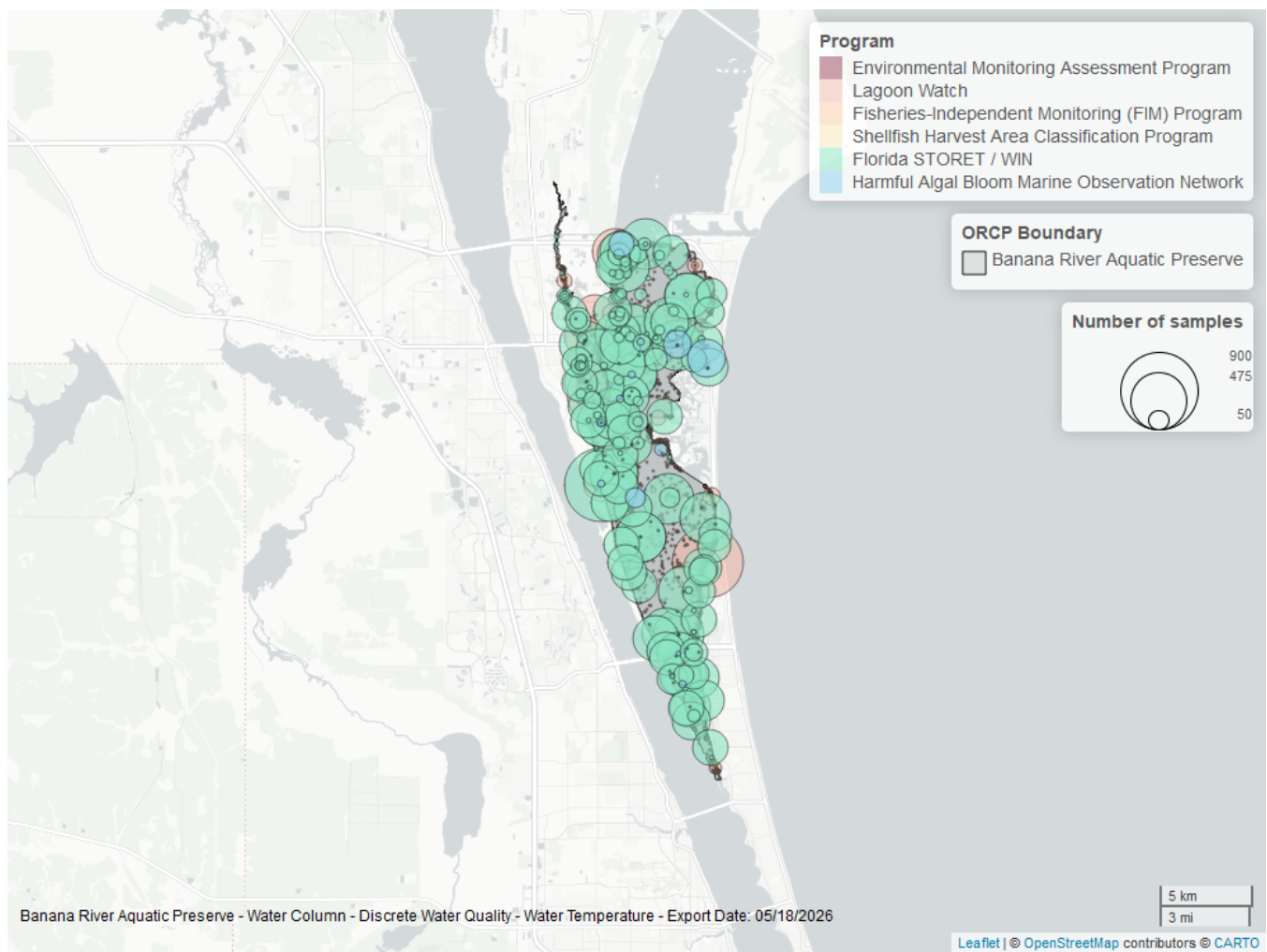


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 27: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
5002	22217	1991	2025
69	4559	1990	2024
3001	4346	1991	2024
95	669	2006	2018
3013	348	2003	2025
540	42	2016	2020
115	7	1995	1995

Program names:

5002 - Florida STORET / WIN¹

69 - Fisheries-Independent Monitoring (FIM) Program⁵

3001 - Lagoon Watch⁶

95 - Harmful Algal Bloom Marine Observation Network⁷

3013 - Indian River Lagoon fixed seagrass transects (SJRWMD)⁸

540 - Shellfish Harvest Area Classification Program²
115 - Environmental Monitoring Assessment Program⁹

Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in Banana River Aquatic Preserve

Table 28: Station overview for Continuous parameters by Program

<i>ProgramID</i>	<i>ProgramLocationID</i>	<i>Years of Data</i>	<i>Use in Analysis</i>	<i>Parameters</i>
5061	CMMerritt	3	FALSE	Turbidity
5061	CMMerritt	4	FALSE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Water Temperature , Salinity , Specific Conductivity
5061	IRLB04	10	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity

Program names:

5061 - St. Johns River Water Management District Continuous Water Quality Programs¹⁰

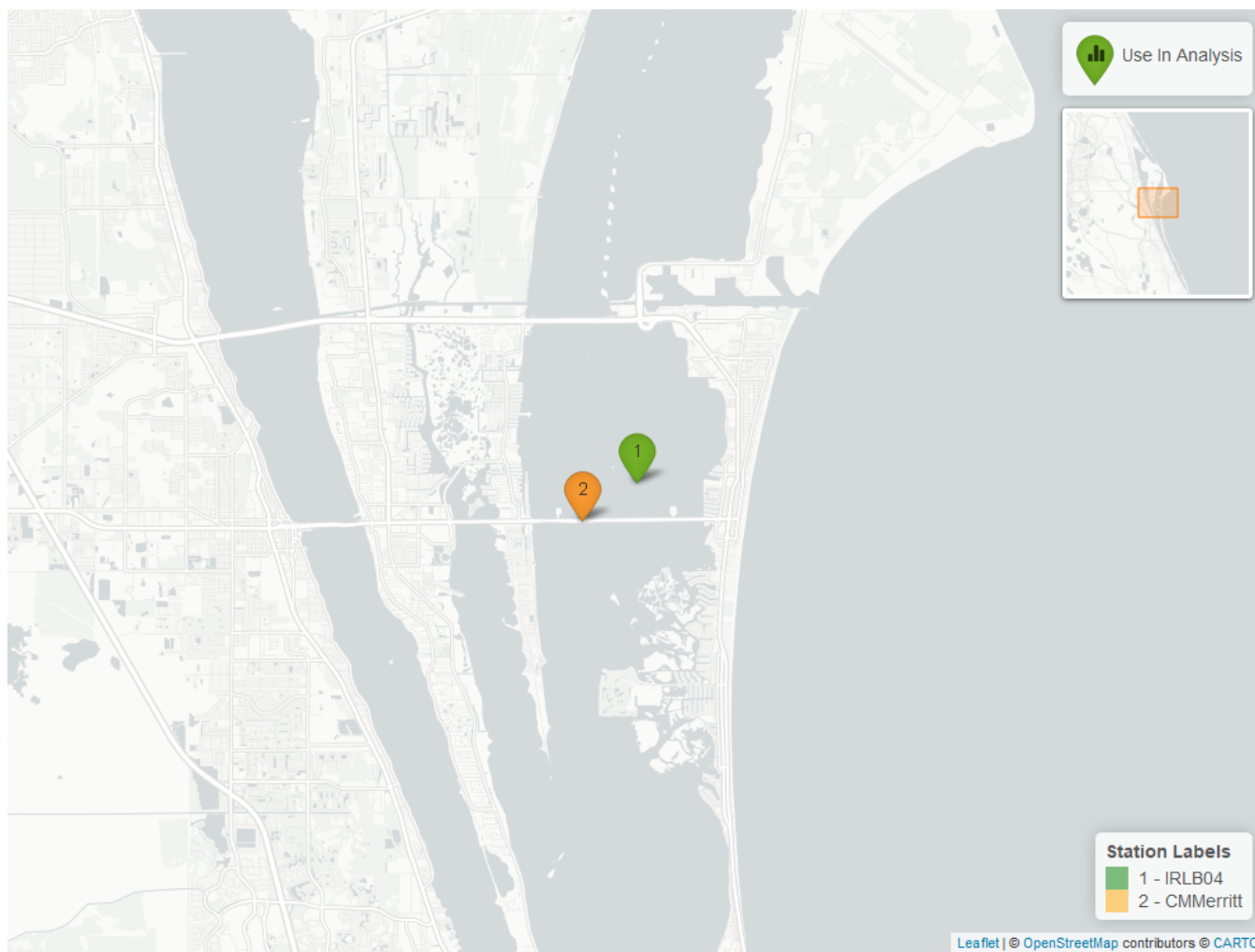


Figure 23: Map showing continuous water quality sampling locations within the boundaries of *Banana River Aquatic Preserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

pH - Continuous

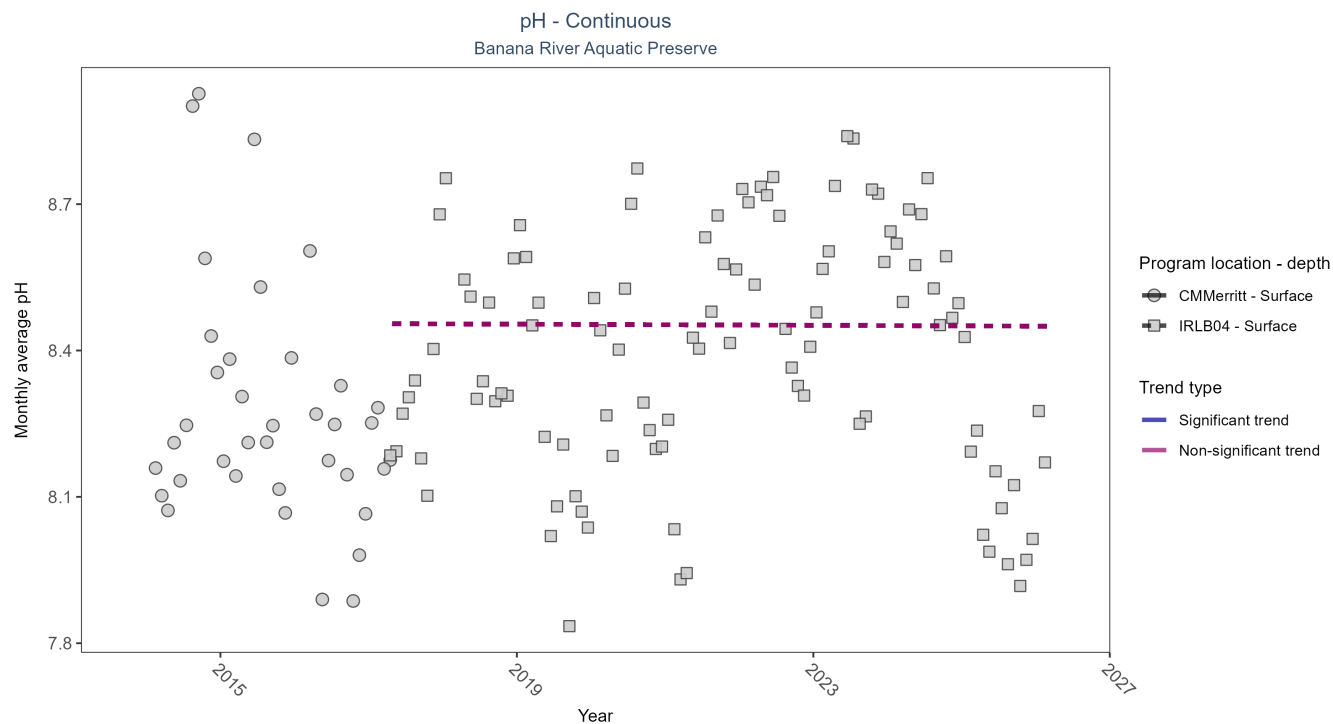


Figure 24: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for pH - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CMMerritt	Insufficient data to calculate trend	27417	4	2014 - 2017	8.22	-	-	-	-
IRLB04	No significant trend	77069	10	2017 - 2026	8.43	-0.01	8.46	0	0.93

No detectable change in monthly average pH was observed at one location. There was insufficient data to fit a model for one location.

Dissolved Oxygen Saturation - Continuous

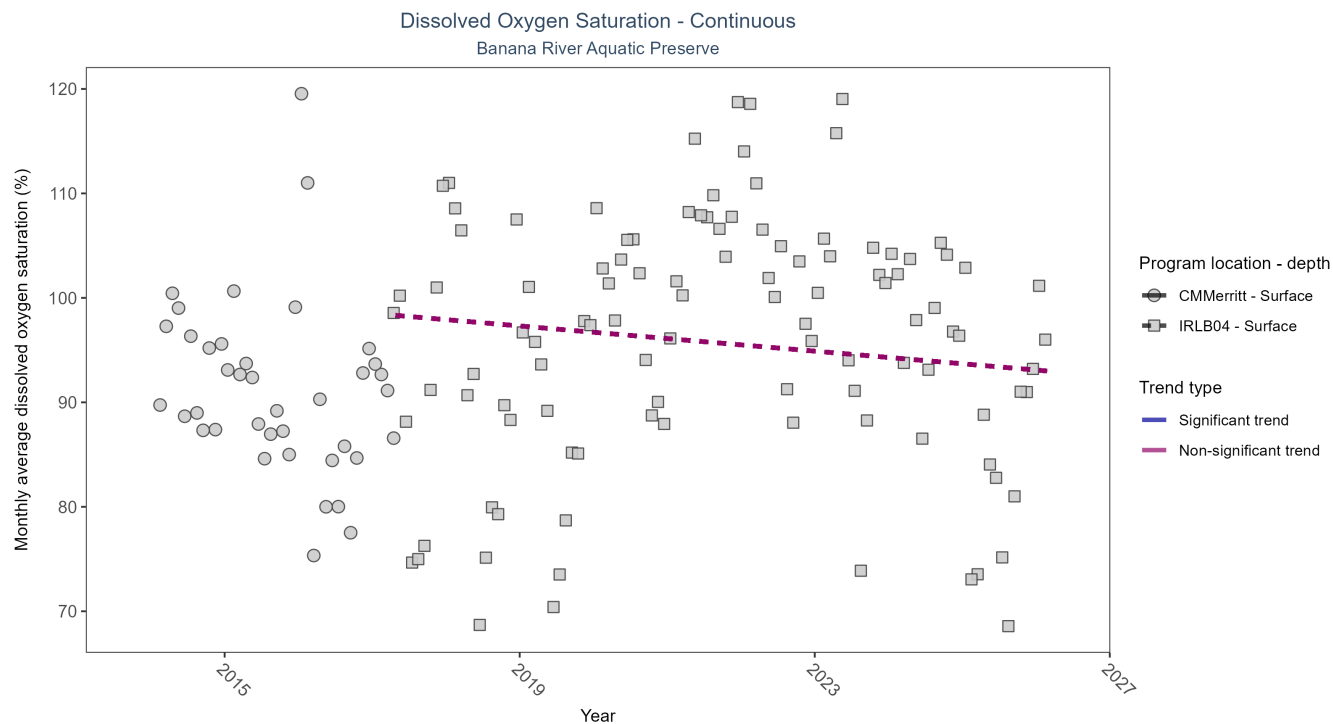


Figure 25: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Dissolved Oxygen Saturation - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CMMerritt	Insufficient data to calculate trend	25864	4	2014 - 2017	91.03	-	-	-	-
IRLB04	No significant trend	89829	10	2017 - 2026	100.13	-0.08	98.52	-0.6	0.31

No detectable change in monthly average dissolved oxygen saturation was observed at one location. There was insufficient data to fit a model for one location.

Dissolved Oxygen - Continuous

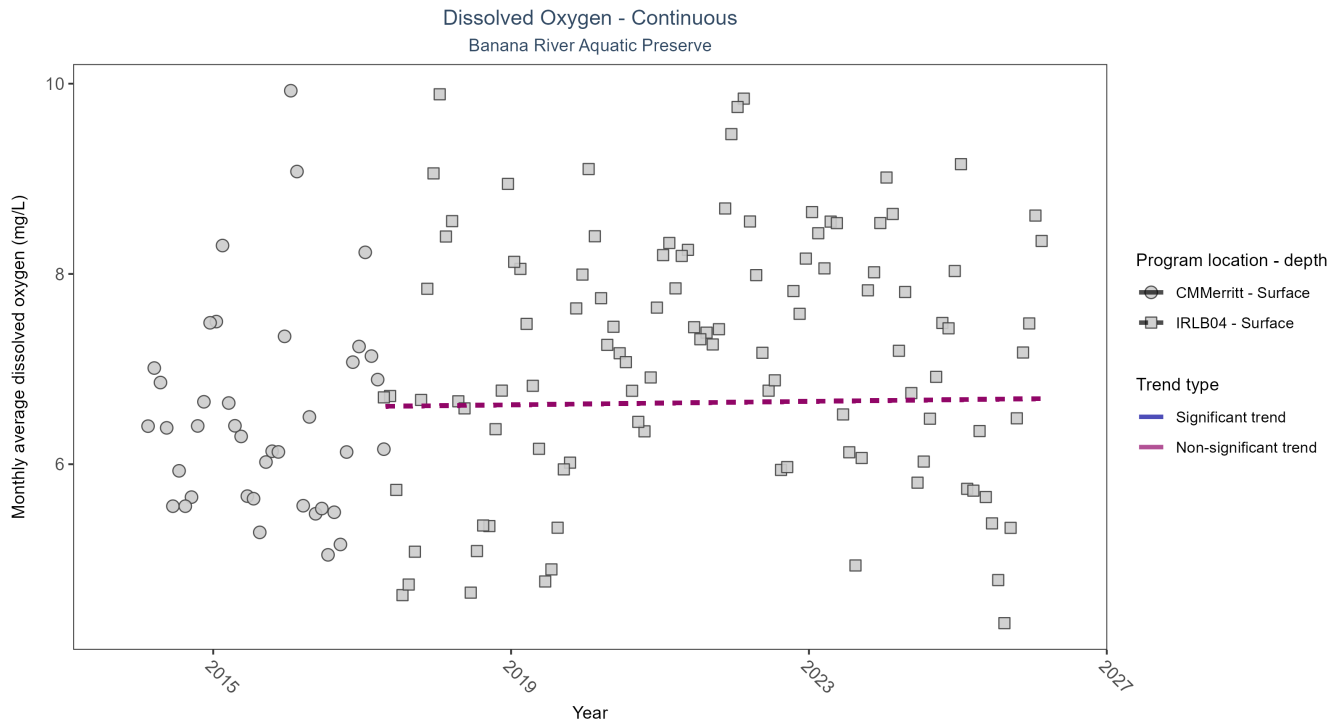


Figure 26: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 31: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CMMerritt	Insufficient data to calculate trend	27378	4	2014 - 2017	6.51	-	-	-	-
IRLB04	No significant trend	77153	10	2017 - 2026	7.21	0.03	6.61	0.01	0.69

No detectable change in monthly average dissolved oxygen was observed at one location. There was insufficient data to fit a model for one location.

Turbidity - Continuous

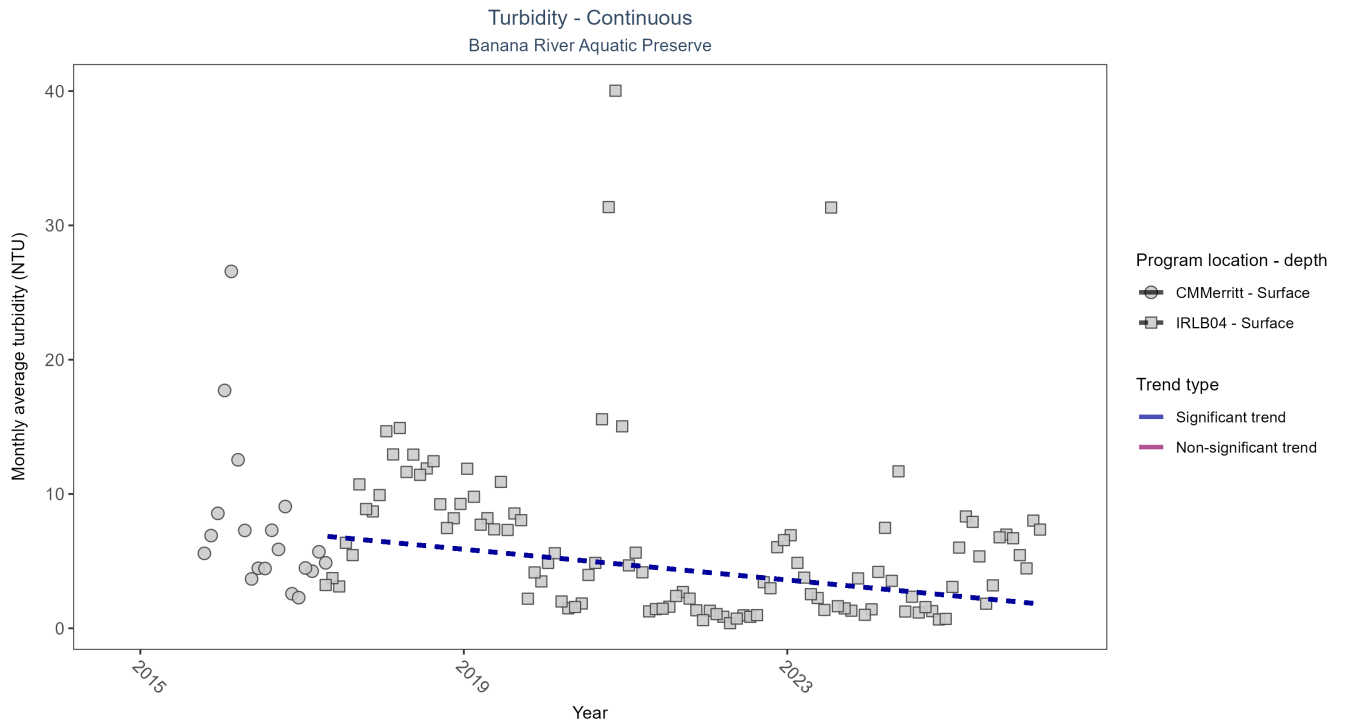


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 32: Seasonal Kendall-Tau Results for Turbidity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CMMerritt	Insufficient data to calculate trend	12912	3	2015 - 2017	5.29	-	-	-	-
IRLB04	Significantly decreasing trend	74218	10	2017 - 2026	3.41	-0.32	7.02	-0.57	0

At one program location, monthly average turbidity decreased by 0.57 NTU per year. There was insufficient data to fit a model for one location.

Water Temperature - Continuous

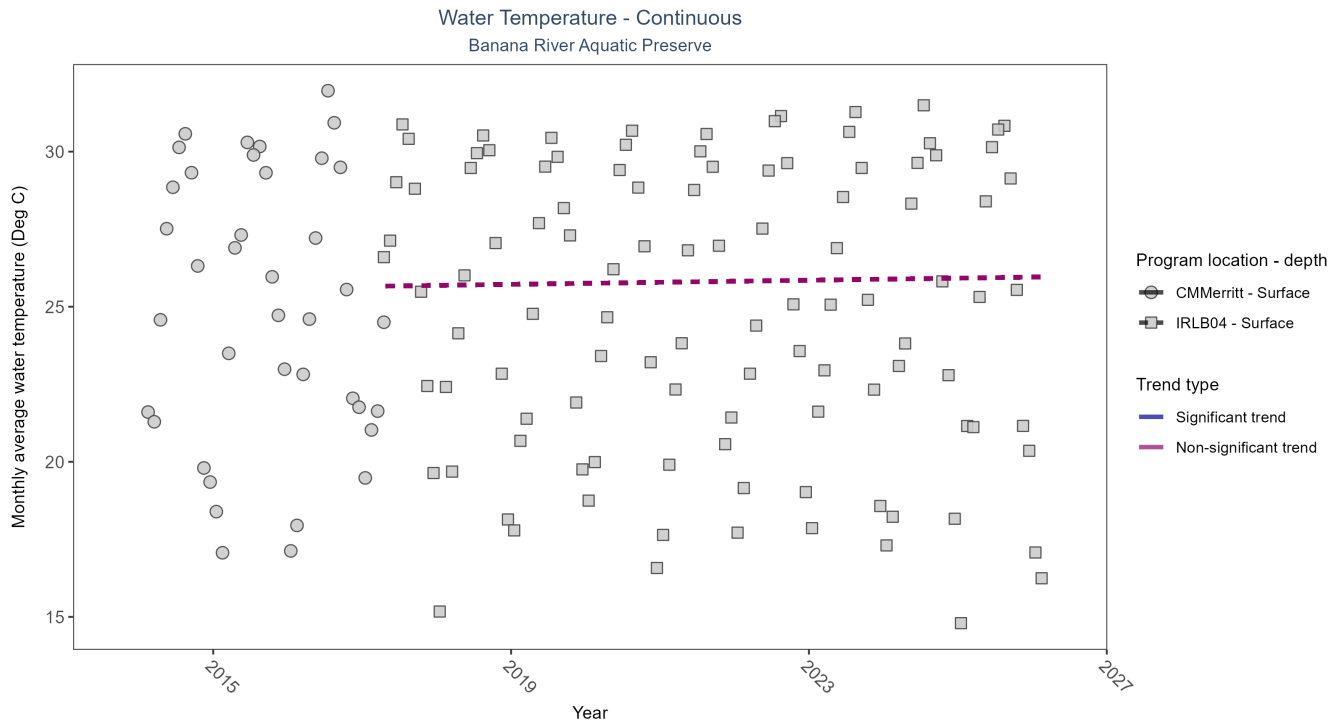


Figure 28: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 33: Seasonal Kendall-Tau Results for Water Temperature - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CMMerritt	Insufficient data to calculate trend	27484	4	2014 - 2017	25.43	-	-	-	-
IRLB04	No significant trend	77157	10	2017 - 2026	25.61	0.05	25.65	0.03	0.52

No detectable change in monthly average water temperature was observed at one location. There was insufficient data to fit a model for one location.

Salinity - Continuous

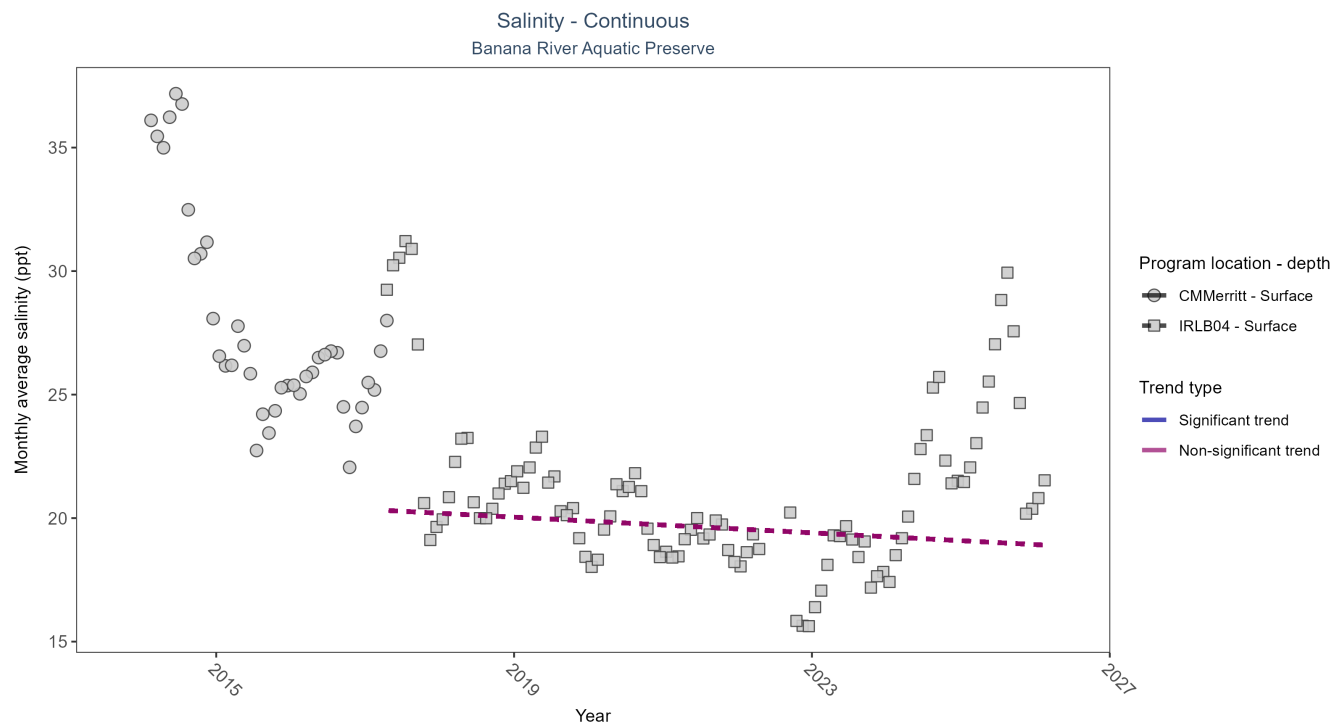


Figure 29: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 34: Seasonal Kendall-Tau Results for Salinity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CMMerritt	Insufficient data to calculate trend	25902	4	2014 - 2017	26.40	-	-	-	-
IRLB04	No significant trend	73363	10	2017 - 2026	20.28	-0.1	20.36	-0.16	0.21

No detectable change in monthly average salinity was observed at one location. There was insufficient data to fit a model for one location.

Specific Conductivity - Continuous

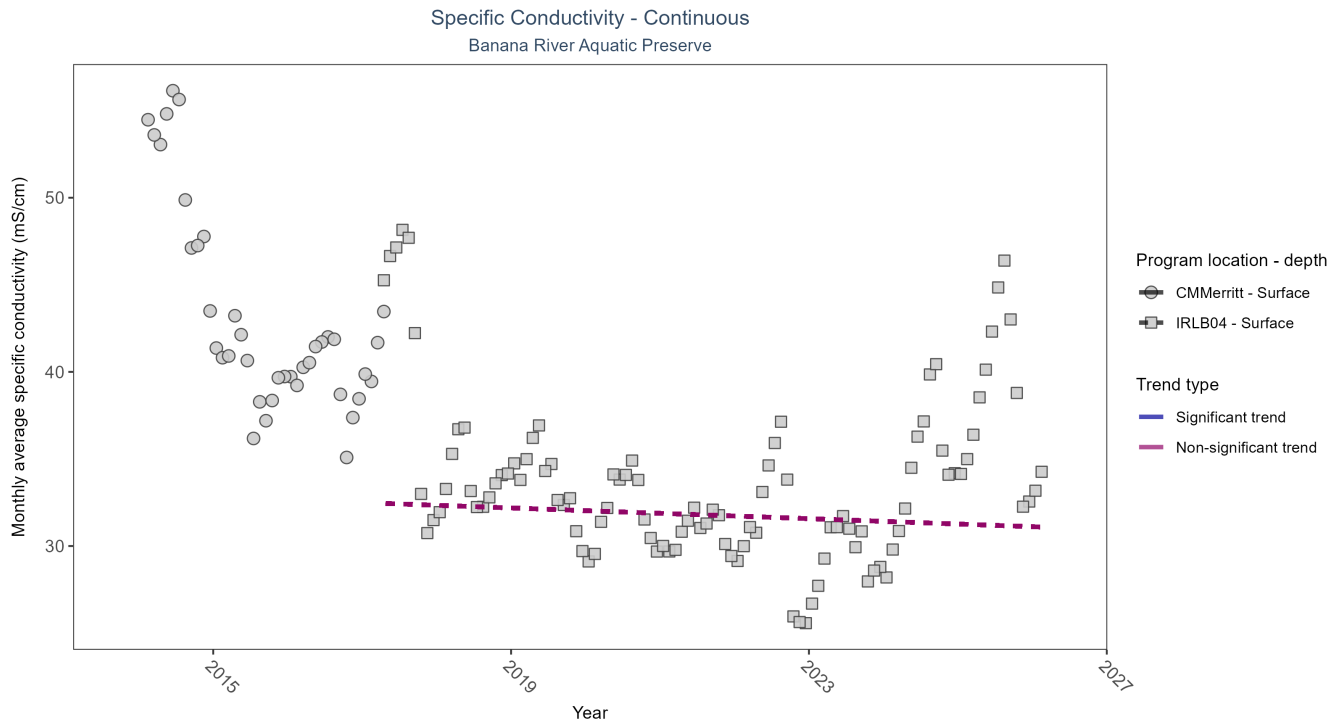


Figure 30: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 35: Seasonal Kendall-Tau Results for Specific Conductivity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
CMMerritt	Insufficient data to calculate trend	25904	4	2014 - 2017	41.33	-	-	-	-
IRLB04	No significant trend	76965	10	2017 - 2026	32.70	-0.06	32.49	-0.15	0.45

No detectable change in monthly average specific conductivity was observed at one location. There was insufficient data to fit a model for one location.

Submerged Aquatic Vegetation

The data file used is: **All_SAV_Parameters-2026-Jun-04.txt**

Submerged aquatic vegetation (SAV) refers to plants and plant-like macroalgae species that live entirely underwater. The two primary categories of SAV inhabiting Florida estuaries are *benthic macroalgae* and *seagrasses*. They often grow together in dense beds or meadows that carpet the seafloor. *Macroalgae* include multicellular species of green, red and brown algae that often live attached to the substrate by a holdfast. They tend to grow quickly and can tolerate relatively high nutrient levels, making them a threat to seagrasses and other benthic habitats in areas with poor water quality. In contrast, *seagrasses* are grass-like, vascular, flowering plants that are attached to the seafloor by extensive root systems. *Seagrasses* occur throughout the coastal areas of Florida, including protected bays and lagoons as well as deeper offshore waters on the continental shelf. *Seagrasses* have taken advantage of the broad, shallow shelf and clear water to produce two of the most extensive seagrass beds anywhere in continental North America.

Parameters

Percent Cover measures the fraction of an area of seafloor that is covered by SAV, usually estimated by evaluating multiple small areas of seafloor. Percent cover is often estimated for total SAV, individual types of vegetation (seagrass, attached algae, drift algae) and individual species.

Frequency of Occurrence was calculated as the number of times a taxon was observed in a year divided by the number of sampling events, multiplied by 100. Analysis is conducted at the quadrat level and is inclusive of all quadrats (i.e., quadrats evaluated using Braun-Blanquet, modified Braun-Blanquet, and percent cover.”

Species

Turtle grass (*Thalassia testudinum*) is the largest of the Florida seagrasses, with longer, thicker blades and deeper root structures than any of the other seagrasses. It is considered a climax seagrass species.

Shoal grass (*Halodule wrightii*) is an early colonizer of vegetated areas and usually grows in water too shallow for other species except *widgeon grass*. It can often tolerate larger salinity ranges than other seagrass species. *Shoal grass* is characterized by thin, flat blades, that are narrower than *turtle grass* blades.

Manatee grass (*Syringodium filiforme*) is easily recognizable because its leaves are thin and cylindrical instead of the flat, ribbon-like form shared by many other seagrass species. The leaves can grow up to half a meter in length. *Manatee grass* is usually found in mixed seagrass beds or small, dense monospecific patches.

Widgeon grass (*Ruppia maritima*) grows in both fresh and salt water and is widely distributed throughout Florida's estuaries in less saline areas, particularly in inlets along the east coast. This species resembles *shoal grass* in certain environments but can be identified by the pointed tips of its leaves.

Three species of *Halophila* spp. are found in Florida - **Star grass** (*Halophila engelmannii*), **Paddle grass** (*Halophila decipiens*), and **Johnson's seagrass** (*Halophila johnsonii*). These are smaller, more fragile seagrasses than other Florida species and are considered ephemeral. They grow along a single long rhizome, with short blades. These species are not well-studied, although surveys are underway to define their ecological roles.

Notes

Star grass, *Paddle grass*, and *Johnson's seagrass* will be grouped together and listed as **Halophila spp.** in the following managed areas. This is because several surveys did not specify to the species level:

- Banana River Aquatic Preserve
- Indian River-Malabar to Vero Beach Aquatic Preserve
- Indian River-Vero Beach to Ft. Pierce Aquatic Preserve
- Jensen Beach to Jupiter Inlet Aquatic Preserve
- Loxahatchee River-Lake Worth Creek Aquatic Preserve
- Mosquito Lagoon Aquatic Preserve

- Biscayne Bay Aquatic Preserve
- Florida Keys National Marine Sanctuary

Banana River Aquatic Preserve
SAV Percent Cover - Sample Locations

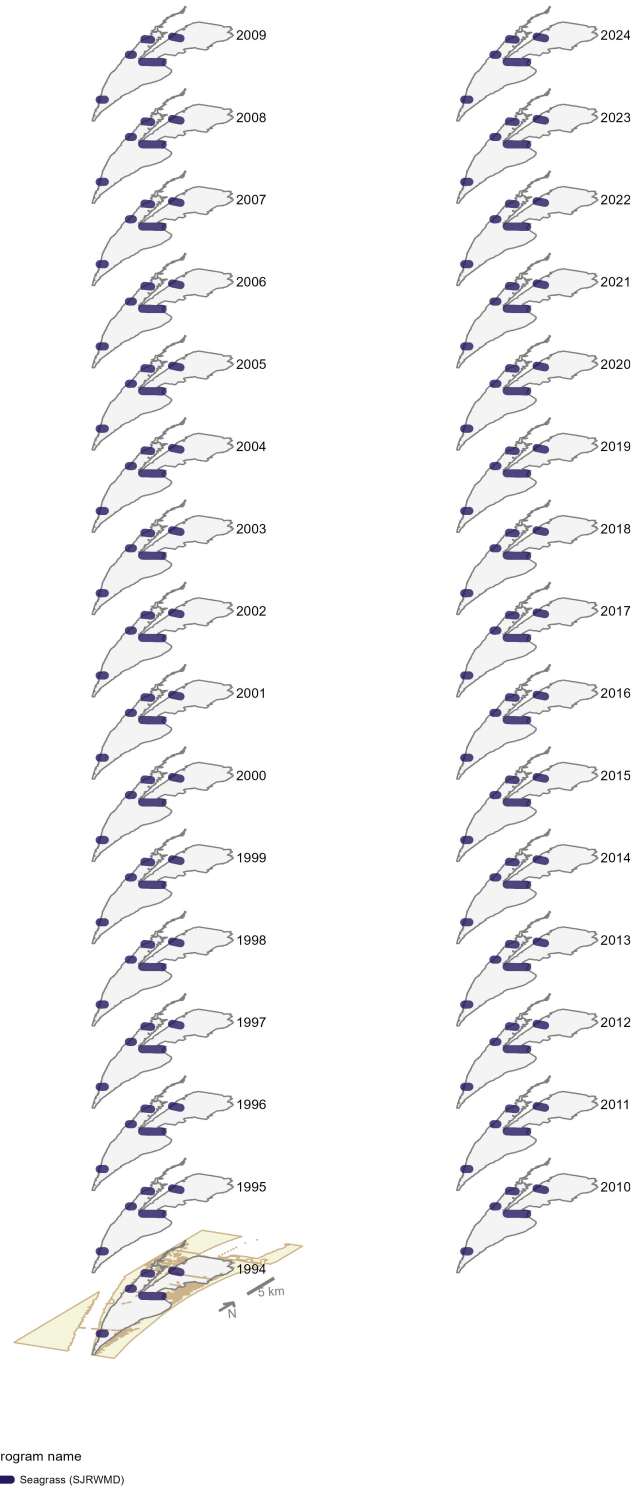


Figure 31: Maps showing the temporal scope of SAV sampling sites within the boundaries of *Banana River Aquatic Preserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

Sampling locations by Program:

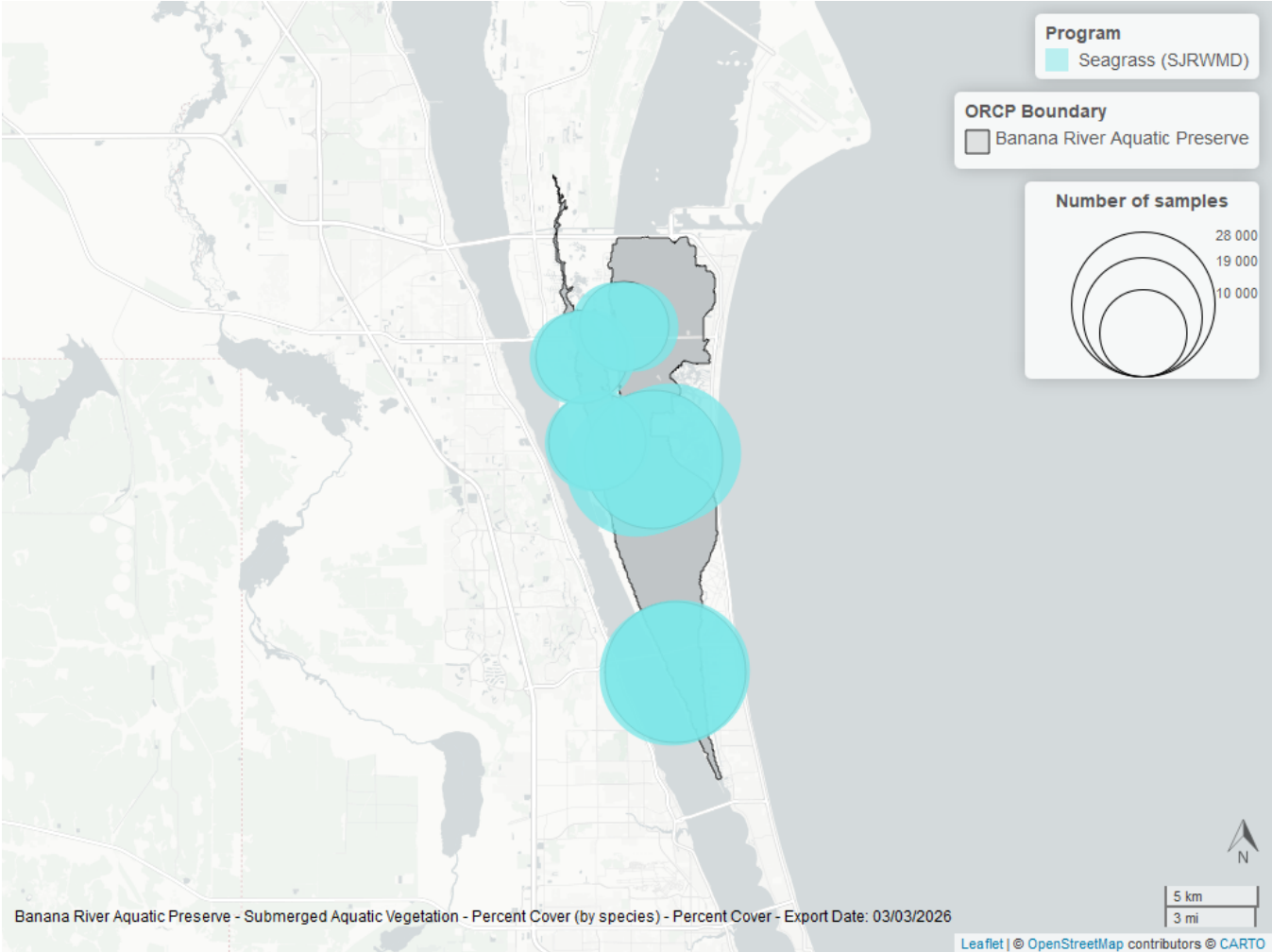


Figure 32: Map showing SAV sampling sites within the boundaries of *Banana River Aquatic Preserve*. The point size reflects the number of samples at a given sampling site.

Table 36: Program Information for Submerged Aquatic Vegetation

<i>ProgramID</i>	<i>N-Data</i>	<i>YearMin</i>	<i>YearMax</i>	<i>method</i>	<i>Sample Locations</i>
3013	41377	1994	2025	Percent Cover	5
3013	47929	1994	2025	Percent Occurrence	5

Program names:

3013 - Seagrass (SJRWMD)⁸

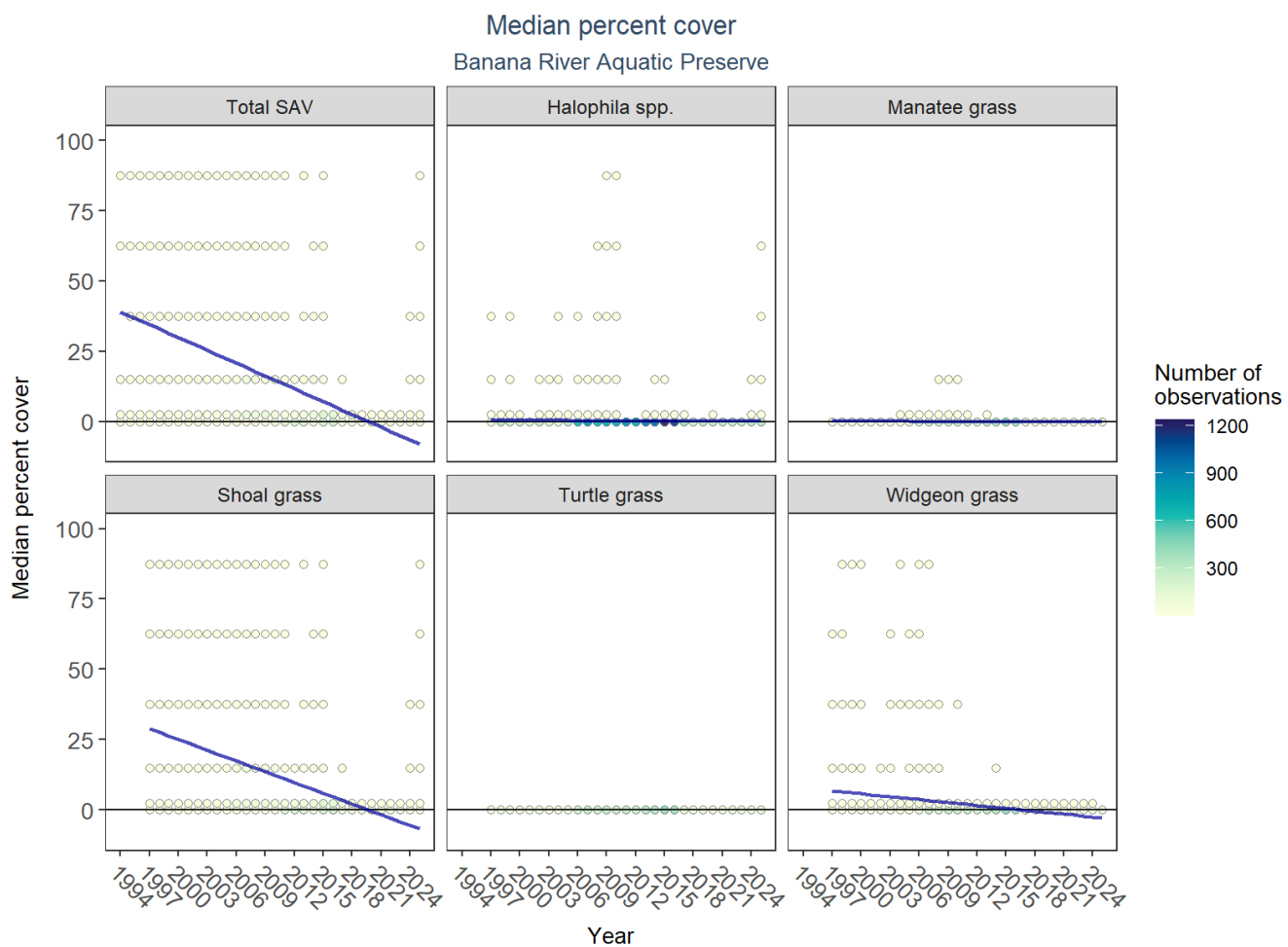


Figure 33: Scatter plots of median percent cover of submerged aquatic vegetation over time by group. Plots for time series that included five or more years of observations show the estimated trend as a blue line.

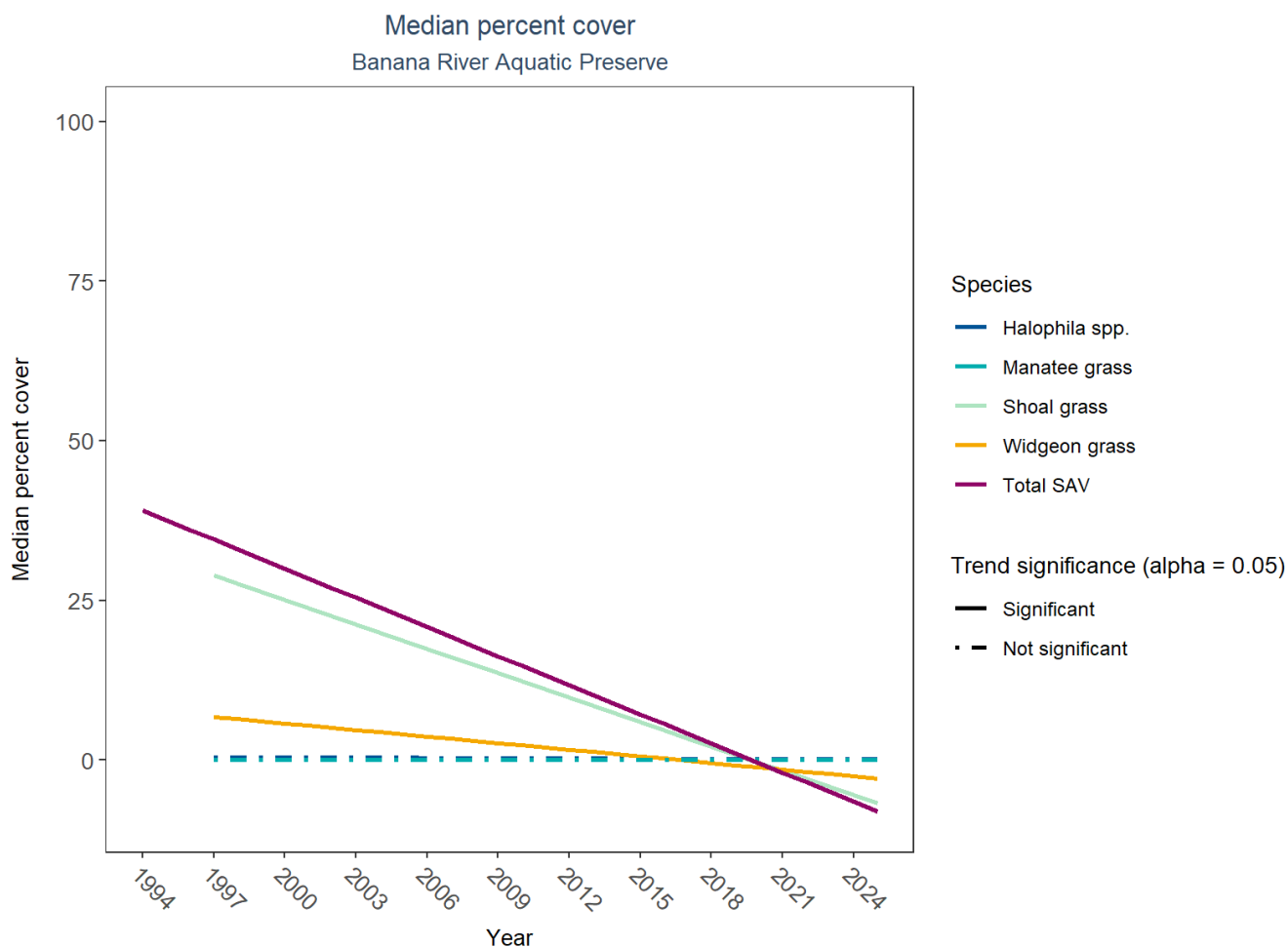


Figure 34: Trend lines of median percent cover of submerged aquatic vegetation over time for species that had five or more years of observations. Line type represents significance (solid) or non-significance (dashed) of the trends.

Table 37: Percent Cover Trend Analysis for Banana River Aquatic Preserve

<i>Species</i>	<i>Statistical Trend</i>	<i>Period of Record</i>	<i>LME Intercept</i>	<i>LME Slope</i>	<i>P</i>
Shoal grass	Decreasing trend	1997 - 2025	32.7066091	-1.2742102	0.0000000
Halophila spp.	No detectable trend	1997 - 2025	0.4997173	-0.0130908	0.1417157
Widgeon grass	Decreasing trend	1997 - 2025	7.7982835	-0.3443284	0.0002304
Manatee grass	No detectable trend	1997 - 2025	0.0924090	-0.0028606	0.0682855
Turtle grass	Model did not fit the available data	1997 - 2025	-	-	-
Total SAV	Decreasing trend	1994 - 2025	39.0719765	-1.5198895	0.0000000

Annual decreases in percent cover were observed for total SAV (-1.52%), shoal grass (-1.27%), and widgeon grass (-0.34%). No detectable change in percent cover was observed for Halophila spp. and manatee grass. The model could not be fitted for turtle grass.

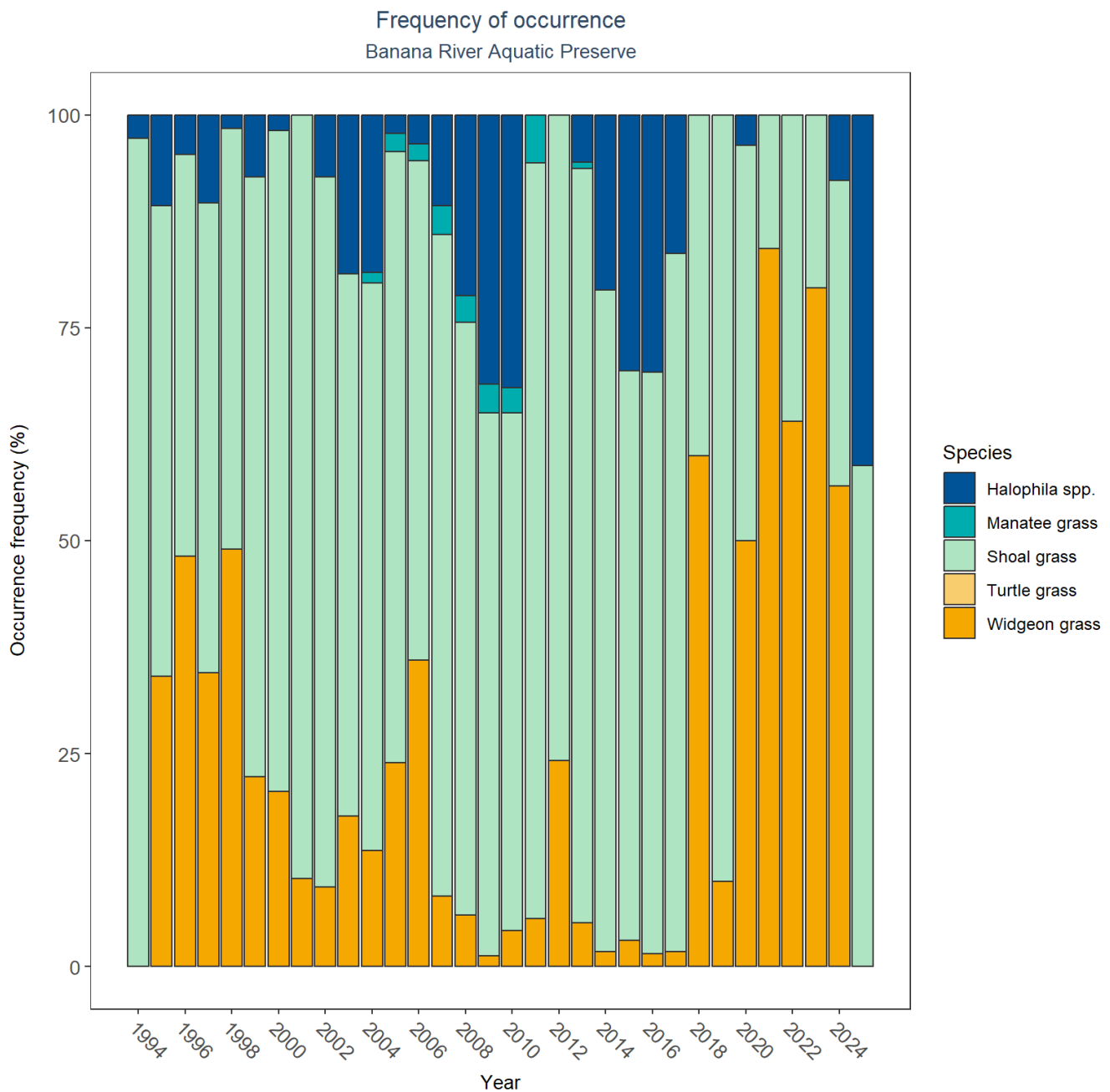


Figure 35: Bar plot of submerged aquatic vegetation species occurrence frequency over time by group.

SAV Water Column Analysis

The following parameters are available for Banana River Aquatic Preserve within the SAV_WC_Report:

- Colored Dissolved Organic Matter
- Chlorophyll a
- Dissolved Oxygen
- Dissolved Oxygen Saturation
- pH
- Salinity

- Secchi Depth
- Water Temperature
- Total Nitrogen
- Total Suspended Solids
- Turbidity

Access the reports here: [DRAFT_SAV_WC_Report.pdf](#)

Species list

Caulerpa ¹	Halophila engelmannii ¹	Thalassia testudinum ¹
Drift algae ¹	Halophila johnsonii ¹	Total SAV ¹
Halodule wrightii ¹	Ruppia maritima ¹	Total seagrass ¹
Halophila decipiens ¹	Syringodium filiforme ¹	Caulerpa ¹

1 - Submerged Aquatic Vegetation

References

1. Florida Department of Environmental Protection (DEP). [Florida STORET / WIN](#). (2024).
2. Florida Department of Agriculture and Consumer Services (FDACS) - Division of Aquaculture. [Shellfish Harvest Area Classification Program](#). (2022).
3. U.S. Environmental Protection Agency (EPA). [EPA STOrage and RETrieval Data Warehouse \(STORET\)/WQX](#). (2023).
4. U.S. Environmental Protection Agency (EPA); Office of Water; National Oceanic and Atmospheric Administration (NOAA); U.S. Geological Survey (USGS); U.S. Fish and Wildlife Service (USFWS); National Estuary Program (NEP); coastal states. [National Aquatic Resource Surveys, National Coastal Condition Assessment](#). (2021).
5. Florida Fish and Wildlife Conservation Commission (FWC). [Fisheries-Independent Monitoring \(FIM\) Program](#). (2022).
6. Volusia County (Florida); Marine Discovery Center. [Lagoon Watch \(Formerly Marine Discovery Center\)](#). (2023).
7. Florida Fish and Wildlife Conservation Commission (FWC); Florida Fish and Wildlife Research Institute (FWRI). [Harmful Algal Bloom Marine Observation Network](#). (2018).
8. St. Johns River Water Management District (SJRWMD). [Seagrass \(SJRWMD\)](#). (2023).
9. U.S. Environmental Protection Agency (EPA); Office of Research and Development. [Environmental Monitoring Assessment Program](#). (2004).
10. St. Johns River Water Management District (SJRWMD). [St. Johns River Water Management District Continuous Water Quality Programs](#). (2024).